

## 186. Forward Error Correction (FEC) sublayer and Physical Medium Attachment (PMA), type 800GBASE-ER1

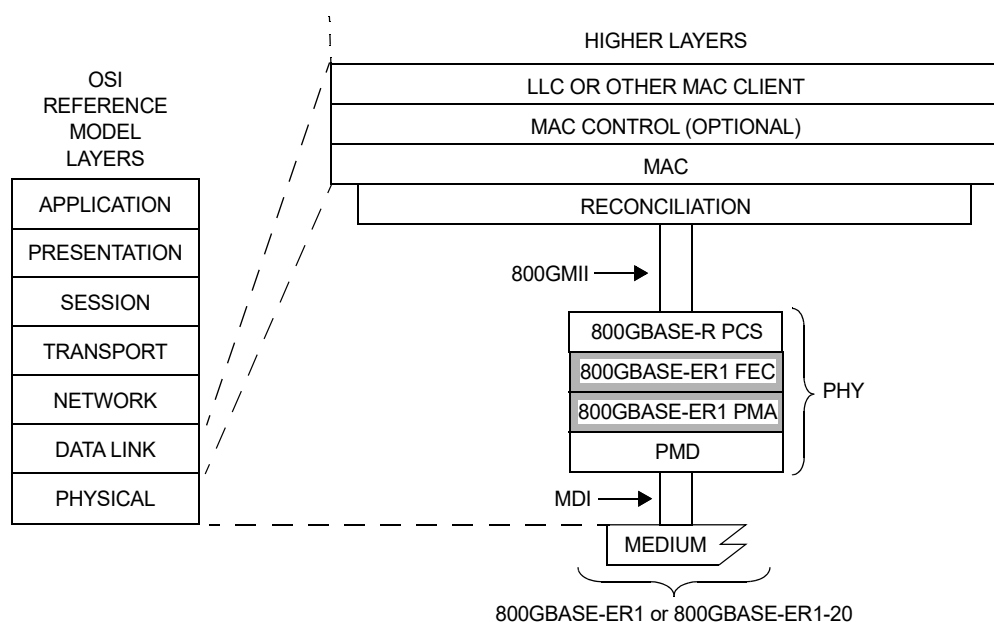
### 186.1 Overview

#### 186.1.1 Scope

This clause specifies the Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer for the Physical Layer implementations known as 800GBASE-ER1-20 and 800GBASE-ER1. The 800GBASE-ER1-20 and 800GBASE-ER1 PHYs (see 187.1) use the 800GBASE-R PCS defined in Clause 172 and the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA defined in this clause.

#### 186.1.2 Relationship of 800GBASE-ER1 FEC and 800GBASE-ER1 PMA to other standards

Figure 186–1 shows the relationship of the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers (shown shaded), along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE  
FEC = FORWARD ERROR CORRECTION SUBLAYER  
MAC = MEDIA ACCESS CONTROL  
MDI = MEDIUM DEPENDENT INTERFACE  
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE  
PMA = PHYSICAL MEDIUM ATTACHMENT  
PMD = PHYSICAL MEDIUM DEPENDENT

**Figure 186–1—800GBASE-ER1 FEC and 800GBASE-ER1 PMA relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model**

#### 186.1.3 Summary of operation

In the transmit direction, the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers together provide RS-FEC decoding, mapping, ER1 FEC encoding, and generation of dual polarization, 16-state quadrature amplitude modulation (DP-16QAM) symbols at the 800GBASE-ER1-20 or 800GBASE-ER1 PMD service

interface. In the receive direction, the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers together decode DP-16QAM symbols from the 800GBASE-ER1-20 or 800GBASE-ER1 PMD service interface, perform ER1 FEC error detection, correction, demapping and decoding, and RS-FEC encoding at the 800GBASE-ER1 FEC service interface. A high-level block diagram of the 800GBASE-ER1 FEC and the 800GBASE-ER1 PMA is shown in Figure 186–2.

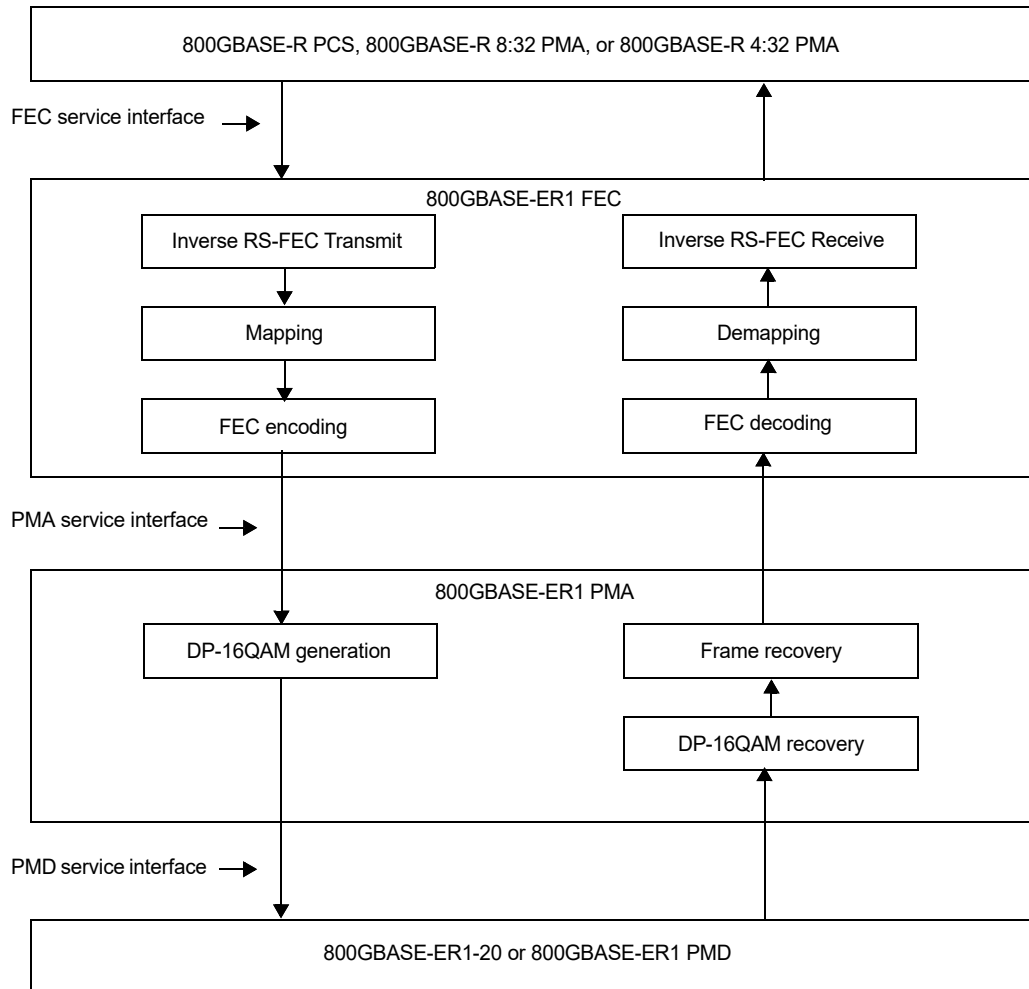


Figure 186–2—800GBASE-ER1 FEC and 800GBASE-ER1 PMA high level block diagram

## 186.2 Forward Error Correction sublayer (FEC), type 800GBASE-ER1

### 186.2.1 800GBASE-ER1 FEC overview

A functional block diagram of the 800GBASE-ER1 FEC sublayer is shown in Figure 186–3.

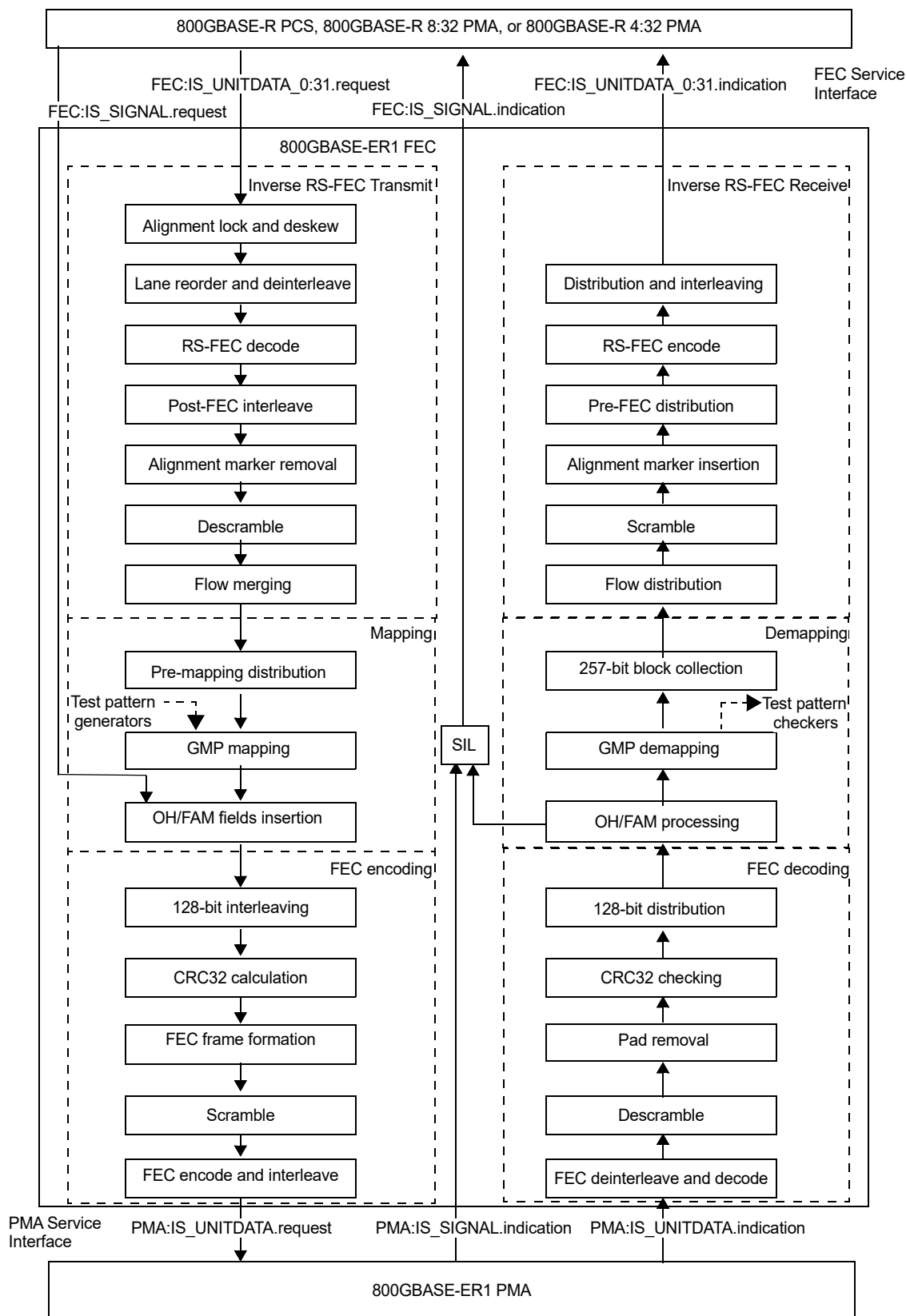


Figure 186-3—800GBASE-ER1 FEC functional block diagram

The dashed lines in Figure 186–3 relate to the high-level functions shown in Figure 186–2.

The 800GBASE-ER1 FEC sublayer is composed of transmit and receive functions. The 800GBASE-ER1 FEC sublayer decodes the RS-FEC that the 800GBASE-R PCS uses and adds a stronger FEC (ER1 FEC) to mitigate transmission errors over the medium.

The 800GBASE-ER1 FEC sublayer communicates with the client sublayer using 32 PCS lanes operating at a nominal rate of 26.5625 Gb/s. For communication with the 800GBASE-ER1 PMA in the transmit direction, the 800GBASE-ER1 FEC sublayer provides ER1 FEC codewords (see 186.2.3.10) to the 800GBASE-ER1 PMA. For communication with the 800GBASE-ER1 PMA in the receive direction, the 800GBASE-ER1 FEC sublayer receives ER1 FEC codewords (see 186.3.2) from the 800GBASE-ER1 PMA.

The 800GBASE-ER1 FEC sublayer transmit function operates in normal mode or test-pattern mode.

When the transmit function is in normal mode, it receives 32 PCS lanes in the form of two PCS flows with 16 PCS lanes each. It aligns and reorders the 32 PCS lanes to recover the two PCS flows. For each PCS flow, it decodes the RS(544,514) FEC, removes alignment markers, descrambles the blocks, and interleaves the PCS lanes to form a serial stream of 257-bit blocks. The 257-bit blocks from the two PCS flows are then merged on a 257-bit block basis to form a single stream of 257-bit blocks. Those blocks are distributed to 8 FEC flows, each of which is mapped into a tributary slot within the 800GBASE-ER1 frame (see 186.2.3.3). Frame alignment and other overhead elements are added to the 800GBASE-ER1 frames. The 8 FEC flows are interleaved and encoded with an extended BCH(256,239) code. The ER1 FEC codewords are sent to the 800GBASE-ER1 PMA via the PMA:IS\_UNITDATA.request primitive.

When the transmit function is in test-pattern mode, the 800GBASE-ER1 FEC output is generated as specified in 186.2.3.12.

When the receive function is in normal mode, the 800GBASE-ER1 FEC sublayer synchronization process continually monitors the PMA:IS\_SIGNAL.indication primitive. When SIGNAL\_OK parameter of the PMA:IS\_SIGNAL.indication primitive is OK, the 800GBASE-ER1 FEC sublayer synchronization process accepts a stream of ER1 FEC codewords in the form of  $m$ -bit digitized bit streams representing the four components of DP-16QAM symbols via the PMA:IS\_UNITDATA.indication primitive. The ER1 FEC decoder generates error-corrected codewords from the incoming data stream on the 800GBASE-ER1 PMA service interface, which are then descrambled. The pad bits are removed and the CRC checking is performed before the 257-bit blocks are distributed to eight FEC flows. The 800GBASE-ER1 FEC sublayer then removes the frame alignment and overhead elements and passes the data to the demapper (see 186.2.4.8). The 8 FEC flows are then interleaved to a serial stream of 257-bit blocks and then distributed to 2 PCS flows. In each PCS flow, scrambling, alignment marker insertion, and RS-FEC encoding are performed, and finally distribution to 32 PCS lanes.

When the receive function is in test pattern mode, it operates identically to when it is in normal mode, with the exception that the output of the demapper passes each of the eight FEC flows to an associated pattern checker instance, as described in 186.2.4.11.

## 186.2.2 800GBASE-ER1 FEC sublayer service interface

The client sublayer above the 800GBASE-ER1 FEC sublayer (see Figure 186–2) is either the 800GBASE-R PCS, 800GBASE-R 8:32 PMA, or 800GBASE-R 4:32 PMA.

The service interface primitives are summarized as follows:

FEC:IS\_UNITDATA\_*i*.request(tx\_symbol)  
FEC:IS\_UNITDATA\_*i*.indication(rx\_symbol)  
FEC:IS\_SIGNAL.request(SIGNAL\_OK)  
FEC:IS\_SIGNAL.indication(SIGNAL\_OK)

The primitives are defined for  $i = 0$  to 31. In the transmit direction, the 800GBASE-ER1 FEC sublayer receives 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, from the client sublayer above. In the receive direction, the 800GBASE-ER1 FEC sublayer sends 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, to the client sublayer.

The 800GBASE-ER1 FEC sublayer receives signal status information from the client sublayer above using the FEC:IS\_SIGNAL.request primitive and provides signal status information to the client sublayer above using the FEC:IS\_SIGNAL.indication primitive (see Figure 186–3). The signal status received via FEC:IS\_SIGNAL.request is mapped to the MNT bits (see 186.2.3.5.11) as shown in Table 186–1.

**Table 186–1—Mapping FEC:IS\_SIGNAL.request to MNT bit values**

| FEC:IS_SIGNAL.request(SIGNAL_OK) | IFEC_align_status | Transmitted MNT field | Payload                 |
|----------------------------------|-------------------|-----------------------|-------------------------|
| OK                               | true              | 000 (OK)              | FEC:IS_UNITDATA.request |
| OK                               | false             | 001 (READY)           | Scrambled Idle          |
| READY                            | don't care        | 001 (READY)           | Scrambled Idle          |
| IN_PROGRESS                      | don't care        | 010 (IN_PROGRESS)     | Scrambled Idle          |
| FAIL                             | don't care        | 011 (FAIL)            | LF                      |

The SIL function uses information received from the 800GBASE-ER1 PMA via PMA:IS\_SIGNAL.indication and information from the received MNT bits (see 186.2.4.7.6) to determine the SIGNAL\_OK value that is communicated via FEC:IS\_SIGNAL.indication. The mapping is as shown in Table 186–2.

**Table 186–2—Generation of FEC:IS\_SIGNAL.indication**

| PMA:IS_SIGNAL.indication (SIGNAL_OK) | Received MNT field | FEC:IS_SIGNAL.indication (SIGNAL_OK) |
|--------------------------------------|--------------------|--------------------------------------|
| OK                                   | 000 (OK)           | OK                                   |
| OK                                   | 001 (READY)        | READY                                |
| OK                                   | 010 (IN_PROGRESS)  | IN_PROGRESS                          |
| OK                                   | 011 (FAIL)         | FAIL                                 |
| FAIL                                 | any                | FAIL                                 |

When FEC:IS\_SIGNAL.indication(SIGNAL\_OK) is FAIL, the rx\_symbol parameters of the FEC:IS\_UNITDATA\_i.indication primitives are undefined.

## 186.2.3 Transmit function

### 186.2.3.1 Inverse RS-FEC Transmit

The inverse RS-FEC Transmit function decodes the RS(544,514) FEC that was added by the 800GBASE-R PCS and creates a single stream of 257-bit blocks to be mapped into the 800GBASE-ER1 frame.

#### 186.2.3.1.1 Alignment lock and deskew

The Alignment lock and deskew function is identical to that specified in 172.2.5.1.

#### 186.2.3.1.2 Lane reordering and deinterleaving

The Lane reordering and deinterleave function is identical to that specified in 172.2.5.2.

#### 186.2.3.1.3 Reed-Solomon decoder

The Reed-Solomon decoder is identical to that specified in 172.2.5.3.

#### 186.2.3.1.4 Post-FEC interleave

The Post-FEC interleave function is identical to that specified in 172.2.5.4.

#### 186.2.3.1.5 Alignment marker removal

The alignment marker removal function is identical to that specified in 172.2.5.5.

The alignment marker removal function may provide the ability to record the location of the removed alignment markers using the OH field as described in 186.2.3.5.10. The presence of this option is indicated by the assertion of the FEC\_alignment\_marker\_location\_ability status variable (set to 1). When the option is provided, it is enabled by the assertion of the FEC\_alignment\_marker\_location\_enable control variable (set to 1).

The am\_sf<2:0> bits are extracted from the alignment markers in the two flows, logically ORed together, and encoded into the 800GBASE-ER1 tributary frame as specified in 186.2.3.5.11.

#### 186.2.3.1.6 Descrambler

The Descrambler function is identical to that specified in 172.2.5.6.

#### 186.2.3.1.7 Flow merge

Following descrambling, the two flows of 257-bit blocks shall be merged in a round-robin fashion into a single stream of 257-bit blocks, starting with a block from flow 0.

### 186.2.3.2 Pre-mapping distribution

The stream of 257-bit blocks shall be distributed to 8 FEC flows (FEC flows 0 to 7) in a round-robin manner. The distribution shall be such that FEC flows 0, 2, 4, and 6 receive 257-bit blocks that originated from PCS flow 0, and FEC flows 1, 3, 5, and 7 receive 257-bit blocks that originated from PCS flow 1. The GMP mapper (see 186.2.3.3), FAM and pad insertion (see 186.2.3.4), and OH insertion (see 186.2.3.5)

### 186.2.3.3 GMP mapper

GMP is a generic mechanism that uses a sigma/delta distribution algorithm to accommodate an arbitrary bit rate difference between a payload and the space in which it is carried. The principles of the GMP mapper used in the 800GBASE-ER1 FEC sublayer are described in ITU-T G.709 Annex D and ITU-T G.709.1 Annex B. Details of the overhead encoding for 800GBASE-ER1 are described in 186.2.3.5.

|     |     |                                 |     |    |  |           |
|-----|-----|---------------------------------|-----|----|--|-----------|
|     | 0   | FAM                             | pad | OH |  | 5140 bits |
| Row | 1   | Payload (5140 bits) 20×257 bits |     |    |  |           |
|     | 127 | Payload (5140 bits) 20×257 bits |     |    |  |           |

**Figure 186-4—800GBASE-ER1 tributary frame structure**

- 1) The first 480 bits of the 800GBASE-ER1 tributary frame carry the Frame Alignment Mechanism (FAM) field, defined in 186.2.3.4.1.
- 2) The next 480 bits are the pad field, defined in 186.2.3.4.2.
- 3) The next 320 bits carry the OH field, as described in 186.2.3.5.
- 4) The remaining 3860 bits of row 0, plus all bits in rows 1 through 127, comprise the payload area of the frame. When the transmit function is in normal mode, the first 5 bits of the payload area are padding bits (transmitted as zero and ignored upon receipt), and the remaining bits are treated as 2555 blocks of 257 bits each. When the transmit function is in test pattern mode, the entire payload area carries the test pattern data.

For each of the eight 800GBASE-ER1 tributary frames, the payload area of a four frame multi-frame (identified by the two low-order bits of the multi-frame alignment signal, see 186.2.3.5.1) is divided into 10 220 GMP words of 257 bits. Each 257-bit GMP word is filled with either data bits (i.e., the logically serialized stream of 257-bit blocks produced according to 186.2.3.2) or stuff bits, which are transmitted as zero and ignored on receipt.

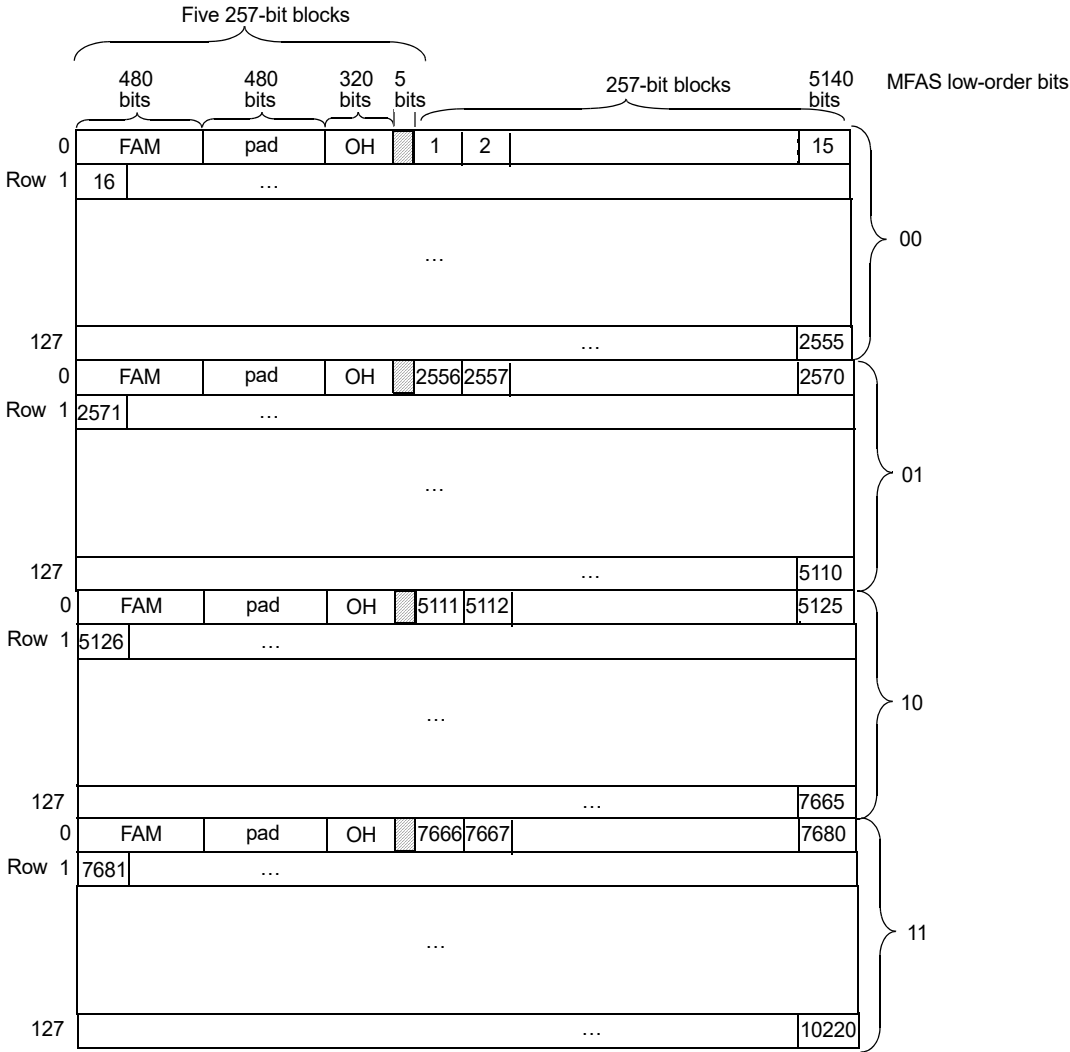


Figure 186–5—800GBASE-ER1 four-frame multi-frame for GMP mapping

The clocks for the encoded 800GBASE-R PCS data and the 800GBASE-ER1 tributary frame may be independent. The nominal rate required for the 257-bit encoded data in each FEC flow is 100.385723 Gb/s  $\pm$  50 ppm (for an aggregate rate of 803.085785 Gb/s  $\pm$  50 ppm). The payload area of the 800GBASE-ER1 tributary frame has a capacity of 100.622438 Gb/s  $\pm$  20 ppm (for an aggregate capacity of 804.979507 Gb/s  $\pm$  20 ppm). The average number of GMP words carrying client data per multi-frame ranges from approximately 10 215.195 to approximately 10 216.625. Since the number of GMP words carrying client data in the multi-frame is always an integer, a given multi-frame will have between 10 215 and 10 217 GMP words that carry data. While the GMP mechanism is generic, the particular clock rates and tolerances result in only three cases, allowing the positions of data and stuff GMP words to be precomputed. The values in Table 186–3 shows the possible locations of GMP stuff words for each possible number of GMP words that are filled with client data.

**Table 186–3—GMP stuff word locations in 800GBASE-ER1 multi-frame**

| Filled GMP words | Stuff GMP word locations (starting from 1) |
|------------------|--------------------------------------------|
| 10 217           | 1, 3407, 6814                              |
| 10 216           | 1, 2556, 5111, 7666                        |
| 10 215           | 1, 2045, 4089, 6133, 8177                  |

#### 186.2.3.4 Frame Alignment Mechanism (FAM) and pad fields

This clause specifies the FAM and pad fields of the 800GBASE-ER1 tributary frame.

##### 186.2.3.4.1 FAM field

The FAM field is used to provide frame delineation for the 800GBASE-ER1 tributary frame. It is inserted before ER1 FEC encoding and removed after ER1 FEC decoding (see Figure 186–3). The content of the FAM field is specified in clause 9.1 of Recommendation ITU-T G.709.6, and consists of 16 bytes of 0x09, 16 bytes of 0xD7, and 28 bytes of 0x00. All bytes of this field are transmitted MSB first. The FAM field is protected by the ER1 FEC. FAM field alignment is processed post-ER1 FEC decode to locate the row corresponding to the start of the 800GBASE-ER1 tributary frame within the payload area of the ER1 FEC frame.

NOTE—This field is named “AM” in Recommendation ITU-T G.709.1, Recommendation ITU-T G.709.6, and OIF-800ZR-01.0.

##### 186.2.3.4.2 Pad

Pad is a 1920-bit all-zeros field. It is ignored upon receipt.

NOTE—This field is named “EOH” in Recommendation ITU-T G.709.1.

##### 186.2.3.5 OH field

The 800GBASE-ER1 tributary frame overhead is present in each of the eight 800GBASE-ER1 tributary frames and is based on the FlexO overhead described in clause 9.2 of Recommendation ITU-T G.709.1. The overhead uses an eight-frame multi-frame, as shown in Figure 186–6 and described in 186.2.3.5.1 through 186.2.3.5.11. The frame overhead is based on the frame defined in subclause 4.3.3 of OIF-800ZR-01.0, which is itself a subset of what is defined in Recommendation ITU-T G.709.1. The 800GBASE-ER1 tributary frame adds additional overhead beyond what is specified in OIF-800ZR-01.0 and Recommendation ITU-T G.709.1 (see 186.2.3.5.10). The bit locations that are not labeled are unused, and are transmitted as zero and ignored by the receiver.

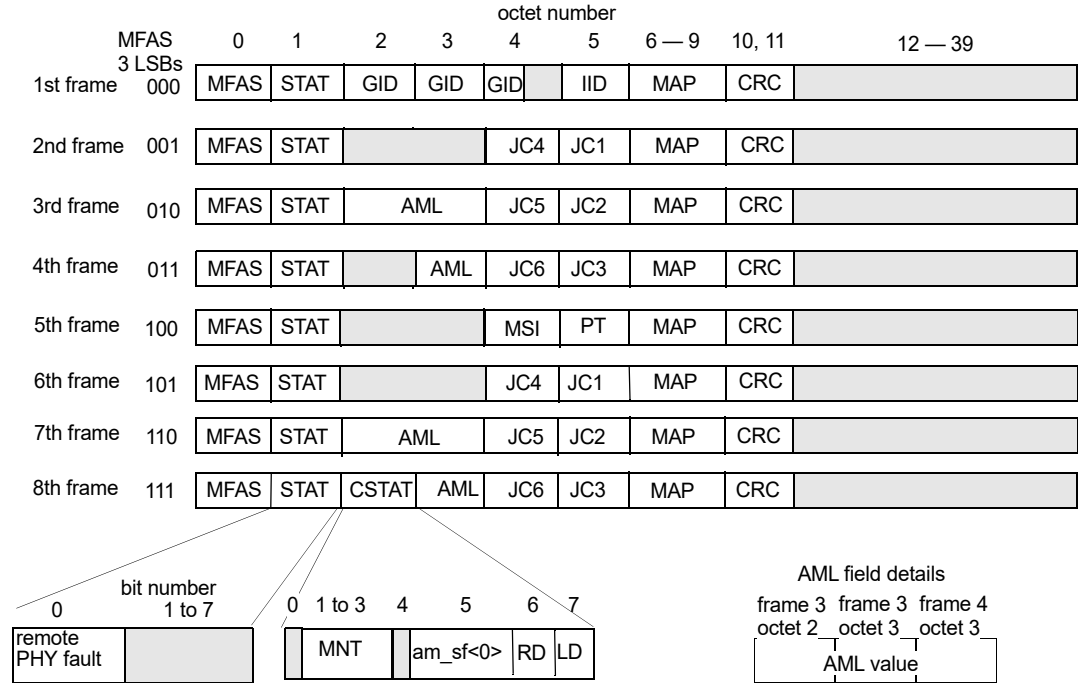


Figure 186–6—Contents of the OH field in eight frame multi-frame

186.2.3.5.1 Multi-frame alignment signal (MFAS)

The multi-frame alignment signal (MFAS) in the first octet of the overhead. As specified in clause 9.2.1 of Recommendation ITU-T G.709.1, it is an auto-wrapping 8-bit counter that is incremented in each 800GBASE-ER1 tributary frame. In the 800GBASE-ER1 FEC sublayer, two multi-frames are used to enable sharing of overhead bytes by information elements that do not need to be repeated every frame (GID, IID, PT, MSI, JC1-JC6, AML) or concatenation of bytes across multiple frames to form a larger field (MAP, AML). The Group Identification (see 186.2.3.5.3), Instance Identification (see 186.2.3.5.4), MAP (see 186.2.3.5.5), Payload Type (see 186.2.3.5.7), and Multiplex Structure Identifier (see 186.2.3.5.8) use an eight-frame multi-frame. The GMP overhead (see 186.2.3.5.9) and Alignment Marker Location overhead (see 186.2.3.5.10) uses a four-frame multi-frame.

186.2.3.5.2 Status octet (STAT)

STAT is present in every frame and provides status information for the 800GBASE-ER1-20 or 800GBASE-ER1 PHY.

The remote PHY fault indication bit is used to signal a remote 800GBASE-ER1 fault. It is set to one by the 800GBASE-ER1 FEC sublayer transmit function to indicate that the local 800GBASE-ER1 FEC sublayer receive function has not detected the location of the FAM field among the stream of 257-bit blocks delivered by the ER1 FEC decoder; otherwise it is set to zero. The remote PHY fault indication bit is in bit 0 of STAT.

186.2.3.5.3 Group ID

The Group ID is a 20-bit field that is specified in clause 9.2.2.1 of Recommendation ITU-T G.709.1. It is present in the first frame of the 8-frame multi-frame. The value is set to 1.

#### 186.2.3.5.4 Instance ID (IID)

The Instance ID is an 8-bit field that is specified in clause 9.2.2.2 of Recommendation ITU-T G.709.1. It is present in the first frame of the 8-frame multi-frame. It is used for the purpose of identifying each 100G frame structure within the group. In the context of the 800GBASE-ER1 FEC sublayer, there are always eight 100G frame structures (i.e., eight 800GBASE-ER1 tributary frames), and the values 1 through 8 are used to identify them.

#### 186.2.3.5.5 MAP

The MAP is a 256-bit field (spread across bytes 6 through 9 of the eight frame multi-frame) that is specified in clause 9.2.2.3 of Recommendation ITU-T G.709.1. It is used to indicate the IIDs that are part of the group. Since the IID values are fixed (see 186.2.3.5.4), the value of this field is also fixed. Byte 6 of the first frame has the value 0x7F, byte 7 of the first frame has the value 0x80, and all other bytes have the value 0x00.

#### 186.2.3.5.6 CRC

The CRC is specified in clause 9.2.5 of Recommendation ITU-T G.709.1. It protects the integrity of bytes 2 through 10 of the overhead field.

#### 186.2.3.5.7 Payload Type (PT)

The payload type is specified in clause 9.2.4.2 of Recommendation ITU-T G.709.1. When the 800GBASE-ER1 FEC sublayer is in normal mode, the value 0x42 is used. When the 800GBASE-ER1 FEC sublayer is in test-pattern mode, the value 0x41 is used.

#### 186.2.3.5.8 Multiplex Structure Identifier (MSI)

The multiplex structure identifier is specified in clause 10.2.3 of Recommendation ITU-T G.709.1. When the 800GBASE-ER1 FEC sublayer is in normal mode, the value 0x80 is used. When the 800GBASE-ER1 FEC sublayer is in test-pattern mode, the value is 0x00.

#### 186.2.3.5.9 GMP justification control overhead (JC1–JC6)

The justification control information is specified in clause 10.2.3 of Recommendation ITU-T G.709.1 and is carried in octets 4 and 5 of the second, third, and fourth frames of the four-frame multi-frame, as shown in Figure 186–6.

When the 800GBASE-ER1 FEC sublayer is in normal mode, the transmit path GMP mapper generates justification control octets (JC1–JC6) which are then used by the receive path GMP demapper to recover the 257-bit data blocks and retune them to the same clock as the remote 800GBASE-R transmit clock. JC1–JC6 are carried once per 4-frame 800GBASE-ER1 tributary frame multi-frame. Because each 800GBASE-ER1 tributary is created from a common clock, the GMP value of the JC1-JC6 overhead is the same in all eight of the 800GBASE-ER1 tributary frames.

There are two parameters that are encoded into the overhead:  $C_m(t)$  indicates the number of 257-bit GMP data words that will be transmitted during the next 800GBASE-ER1 FEC multi-frame, while  $\Sigma C_{nD}(t)$  nominally indicates the running remainder. The long-term average value of  $C_m(t) + \Sigma C_{nD}(t)$  represents the incoming serial stream rate as the number of information octets arriving at the GMP encoder per 800GBASE-ER1 tributary frame multi-frame.  $C_m(t)$  is encoded in bits C1–C14 of JC1 and JC2, with the MSB in C1.  $\Sigma C_{nD}(t)$  is encoded in bits D1–D5 of JC4 and JC5 with the MSB in D1. Refer to Annex B of Recommendation ITU-T G.709.1 for additional information on the encoding of JC1-JC6.

When the 800GBASE-ER1 FEC sublayer is in test-pattern mode, the justification control overhead bytes are transmitted as all zeros (there is no rate adaptation when in test-pattern mode; the test pattern and 800GBASE-ER1 tributary frame are generated from the same clock).

#### 186.2.3.5.10 Alignment marker location control overhead (AML)

The three bytes of the AML field (row 3, octets 2 and 3, and row 4, octet 3) in each multi-frame form a single 24-bit field, as shown in Figure 186–6. This field is used to encode information about the location of 800GBASE-R PCS alignment markers that were removed by the Inverse RS-FEC transmit function (186.2.3.1). The field encodes the position of the first non-stuff block that is mapped into the payload area relative to the location of the 800GBASE-R PCS alignment markers that were removed. This information allows the 800GBASE-R PCS alignment markers to be reinserted in the same location by the 800GBASE-ER1 FEC sublayer receive function.

When the alignment marker location feature is supported as indicated by the FEC status variable `FEC_alignment_marker_location_ability` (set to 1) and enabled by the FEC control variable `FEC_alignment_marker_location_enable` (set to 1), the 800GBASE-ER1 FEC sublayer maintains a counter of 257-bit blocks between consecutive sets of 800GBASE-R PCS alignment markers, with the first 257-bit block after the alignment markers in flow 0 being block 1. The AML field is encoded with the value of the counter for the first non-stuff 257-bit block that is mapped into the payload area of the 800GBASE-ER1 tributary multi-frame. For example, if the first 257-bit block in an 800GBASE-ER1 tributary frame is the ninth 257-bit block after 800GBASE-R PCS alignment markers were removed, the AML field would have the value 9, indicating that the 800GBASE-R PCS alignment markers were removed nine 257-bit blocks prior to that block. This is illustrated in Figure 186–7.

The upper part of the figure shows the stream of 257-bit blocks to be mapped into the 800GBASE-ER1 tributary multi-frame, along with the location of the alignment markers that were removed. The bottom part of the figure shows the 800GBASE-ER1 tributary multi-frame.

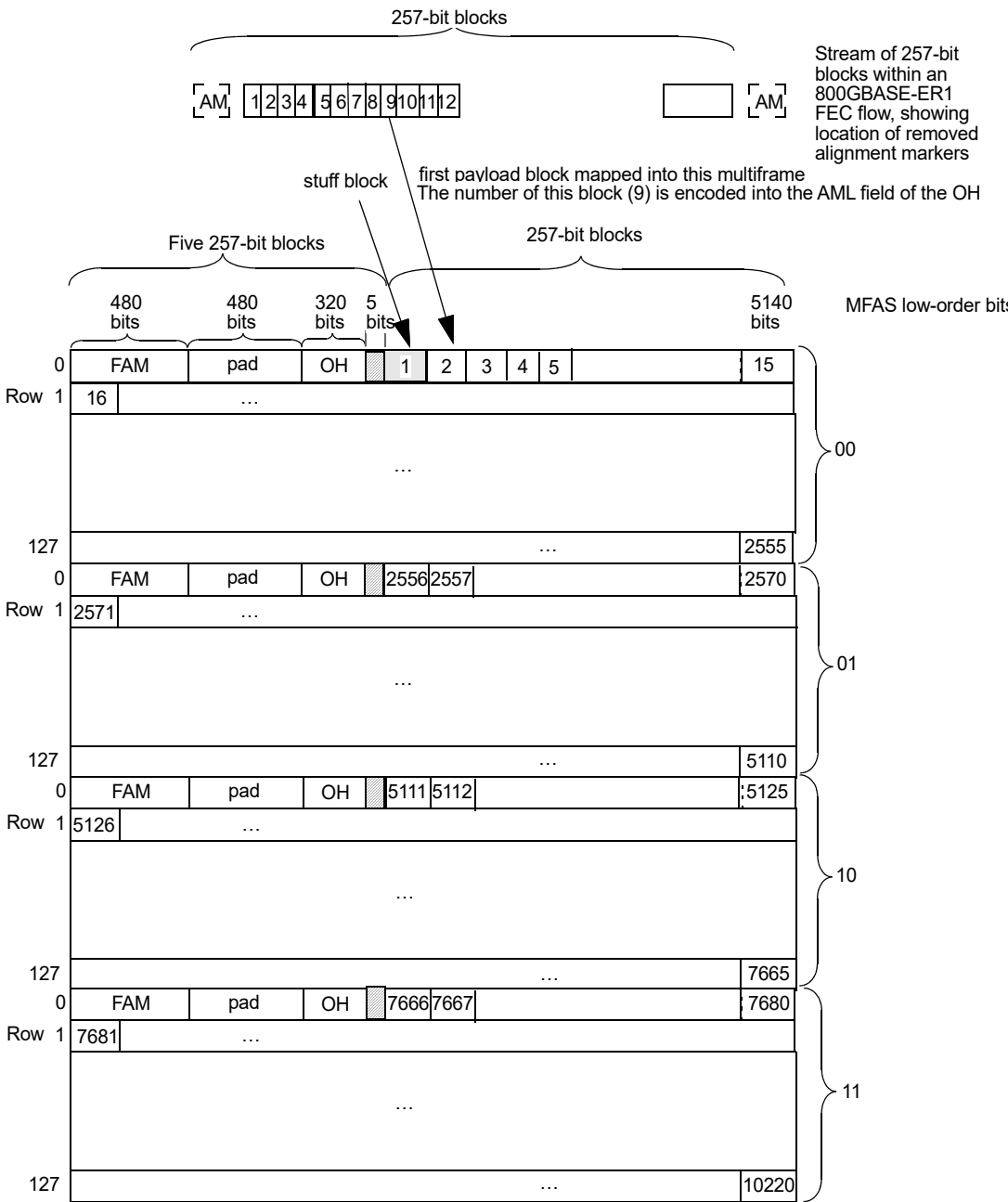


Figure 186–7— Example of AML field usage

If the alignment marker location feature is not supported or not enabled, the value of the AML field is set to all zeros.

186.2.3.5.11 Client status (CSTAT)

The client status information is present in frame 8 of the 8-frame multi-frame. There are multiple pieces of information carried in this field.

Bits 5-7 are used to convey the information encoded in  $\text{am\_sf}\langle 2:0 \rangle$  across the 800GBASE-ER1 or 800GBASE-ER1-20 link according to the following rules:

- The  $\text{am\_sf}\langle 2:0 \rangle$  bits from the two flows are logically ORed together
- The resulting  $\text{am\_sf}\langle 0 \rangle$  signal is placed in  $\text{CSTAT}\langle 5 \rangle$
- The resulting  $\text{am\_sf}\langle 1 \rangle$  signal is logically ORed with  $\text{IFEC\_degraded\_SER}$  and placed in  $\text{CSTAT}\langle 7 \rangle$
- The resulting  $\text{am\_sf}\langle 2 \rangle$  signal is placed in  $\text{CSTAT}\langle 6 \rangle$

Bits 1-3 are the maintenance (MNT) bits and are used to convey the  $\text{SIGNAL\_OK}$  status across the 800GBASE-ER1 or 800GBASE-ER1-20 link as described in Table 186-1.

### 186.2.3.6 128-bit interleaving

The eight 800GBASE-ER1 tributary frames are interleaved on a 128-bit basis in a round-robin manner to form the 800GBASE-ER1 frame. This produces a structure with 128 rows of 41 120 bits. This structure may alternatively be represented as a frame with 512 rows of 10 280 bits. Figure 186-8 shows both of these representations of the 800GBASE-ER1 frame.

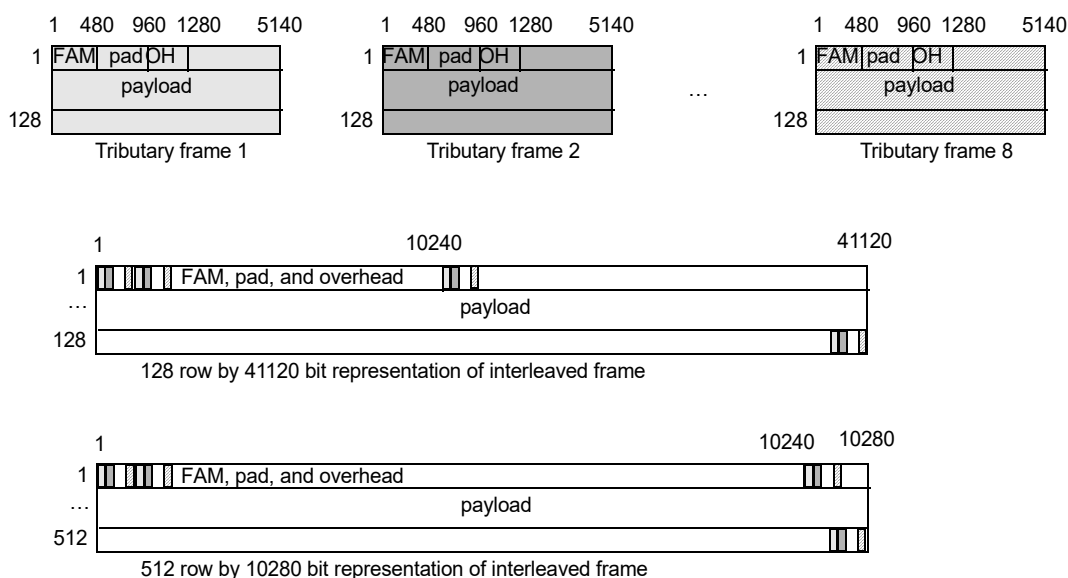


Figure 186-8—128-bit block interleaving of tributary frames

### 186.2.3.7 CRC32 insertion

A 32-bit cyclic redundancy code is calculated over each group of 41 120 bits (i.e., one row of the 128-row representation or four rows of the 512-row representation of the 800GBASE-ER1 frame) as specified in clause 12.4.1 of Recommendation ITU-T G.709.6. The CRC is inserted following the 41 120 bits.

### 186.2.3.8 ER1 FEC frame formation

The ER1 FEC frame is formed from 116 rows of the 512-row representation of the 800GBASE-ER1 frame (1 192 480 bits). Each group of four rows is extended with the CRC32 (see 186.2.3.7). The 29th group of four rows is further extended with a 64-bit pad. The FEC frame consists of 1 193 472 bits. The location of the 800GBASE-ER1 frames within the ER1 FEC frame will vary in a pattern that repeats every 128 ER1

FEC frames. Figure 186–9 illustrates an example of the ER1 FEC frame relative to an 800GBASE-ER1 frame.

NOTE—The ER1 FEC frame in this standard corresponds to the OFBG84 structure defined in Recommendation ITU-T G.709.6.

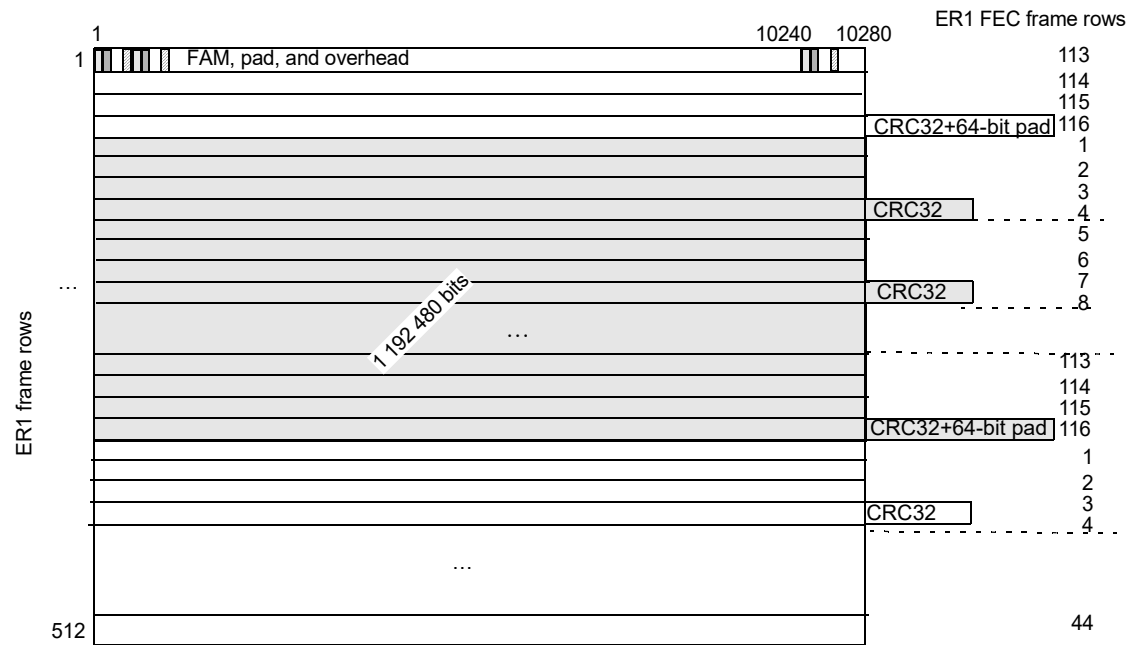


Figure 186–9—ER1 FEC frame formation

186.2.3.9 Scrambler

A frame-synchronous scrambler as specified in clause 12.4.3 of Recommendation ITU-T G.709.6 is applied to the ER1 FEC frame.

186.2.3.10 ER1 FEC encoder and interleaving

The scrambled ER1 FEC frame is encoded as described in clauses 12.4.4 and 12.4.5 of Recommendation ITU-T G.709.6, for the case of FlexO-8e-DO with parameters  $k = 8$  and  $j = 4$ .

The output of the ER1 FEC encoders is interleaved as described in clause 12.4.6 of Recommendation ITU-T G.709.6.

186.2.3.11 Transmit bit ordering and distribution

The transmit bit ordering and distribution for the 800GBASE-ER1 FEC frame shall be as illustrated in Figure 186–10.

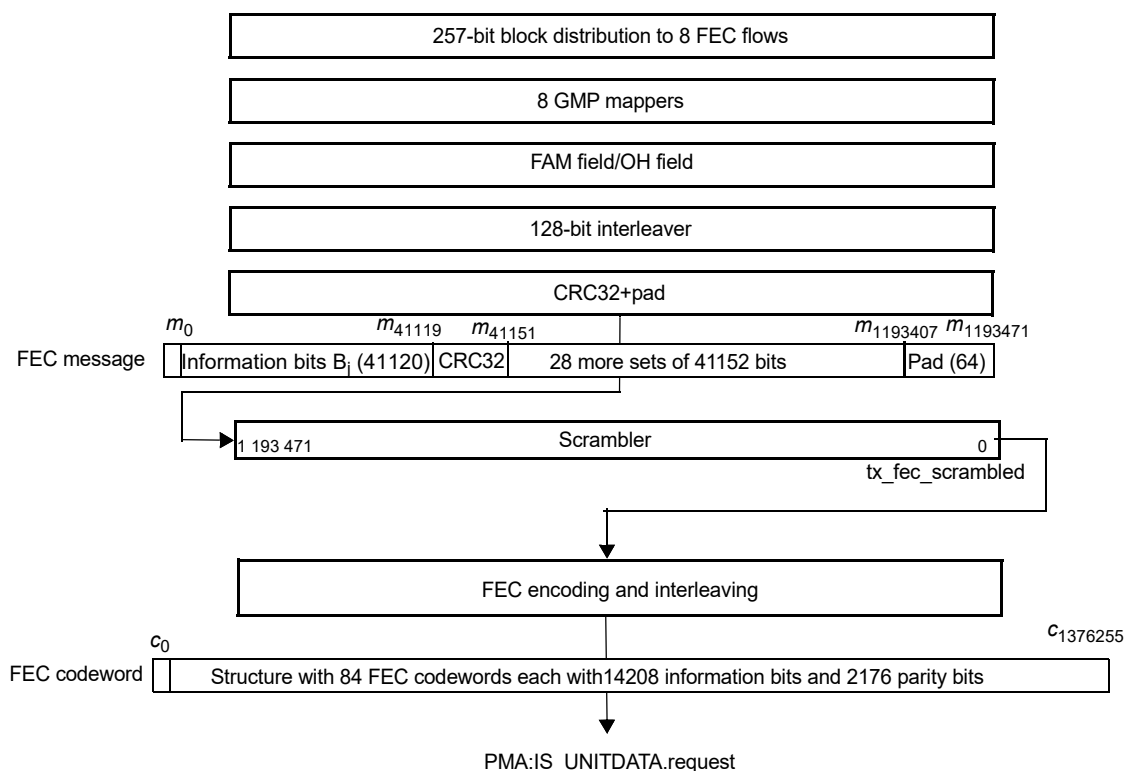


Figure 186-10—800GBASE-ER1 FEC transmit bit ordering and distribution

### 186.2.3.12 Test-pattern generator

The 800GBASE-ER1 FEC sublayer shall have the ability to generate a PRBS31 test pattern, with initial state of all ones. When test pattern generation is enabled by the FEC control variable tx\_test\_mode (set to 1), the output of the test pattern generator is replicated into the payload area of each of the eight 800GBASE-ER1 tributary frames before the GMP mapping process (Figure 186-3), filling the entire payload area as shown in Figure 186-11. The PRBS31 test pattern is as specified in 49.2.8.

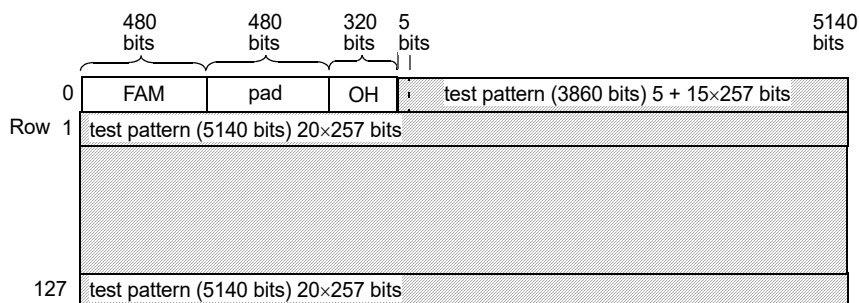


Figure 186-11—Location of test pattern in 800GBASE-ER1 tributary frame

## 186.2.4 Receive function

### 186.2.4.1 ER1 FEC deinterleaving and decoder

The interleaved bits of the ER1 FEC block are deinterleaved by reversing the interleaving process described in 186.2.3.10.

The ER1 FEC decoder shall extract blocks of 1 193 472 bits from an incoming ER1 FEC codeword of 1 376 256 bits. The incoming FEC codeword is formed from a digitized representation of 172 032 DP-16QAM symbols. The incoming DP-16QAM symbols are digitized to an  $m$ -bit resolution by the 800GBASE-ER1 PMA sublayer receive function (see 186.3.4.1) and provided to the 800GBASE-ER1 FEC sublayer receive function by the PMA:IS\_UNITDATA.indication primitive in 186.3.2.

The ER1 FEC decoder is a soft decision decoder and so requires a higher resolution than 2 bits / 4 levels for each of the signals  $X_I$ ,  $X_Q$ ,  $Y_I$ , and  $Y_Q$ . The decoder evaluates the incoming codeword and determines the most likely message value.

The decoded blocks of 1 193 472 bits are passed to the descrambler.

The following counters shall be implemented:

FEC\_corrected\_cw\_counter

The FEC\_corrected\_cw\_counter is a 32-bit counter that counts once for each corrected FEC codeword processed when pma\_alignment\_valid is true. A corrected FEC codeword is a codeword that contained errors and was corrected (including FEC codewords that were miscorrected).

FEC\_uncorrected\_cw\_counter

The FEC\_uncorrected\_cw\_counter is a 32-bit counter that counts once for each uncorrected FEC codeword processed when pma\_alignment\_valid is true. An uncorrected FEC codeword is a codeword that contains errors that were not corrected

FEC\_total\_bits\_counter

The FEC\_total\_bits\_counter is a 64-bit counter that counts once for each bit processed by the FEC decoder when pma\_alignment\_valid is true. This may be used together with the FEC\_corrected\_bits\_counter as a measure of pre-FEC BER.

FEC\_corrected\_bits\_counter

The FEC\_corrected\_bits\_counter is a 64-bit counter that counts once for each bit corrected by the FEC decoder when pma\_alignment\_valid is true. This may be used together with the FEC\_total\_bits\_counter to calculate pre-FEC BER.

### 186.2.4.2 Link degrade signaling

The 800GBASE-ER1 FEC sublayer may provide the ability to signal the presence of a degraded received signal. The presence of this option is indicated by the assertion of the FEC\_degraded\_SER\_ability variable (see 186.7). When the option is provided it is enabled by the assertion of the FEC\_degraded\_SER\_enable variable (see 186.7).

When FEC\_degraded\_SER\_enable is asserted, the 800GBASE-ER1 FEC sublayer computes the number of uncorrected errors detected by ER1 FEC based on the FEC\_uncorrected\_cw\_counter. When the number of uncorrected errors exceeds the threshold set in FEC\_degraded\_SER\_activate\_threshold (see 186.7), the FEC\_degraded\_SER bit (see 186.7) is set. At the end of each interval, if the number of uncorrected errors detected is less than FEC\_degraded\_SER\_deactivate\_threshold, the FEC\_degraded\_SER bit is cleared. If either FEC\_degraded\_SER\_ability or FEC\_degraded\_SER\_enable is deasserted then the FEC\_degraded\_SER bit is cleared.

### 186.2.4.3 Descrambler

The descrambler shall process sequences of 1 193 472 bits to reverse the effects of the scrambler using the polynomial given in 186.2.3.9.

### 186.2.4.4 Remove pad

The descrambled 1 193 472 bits are organized into the ER1 FEC frame described in 186.2.3.8, and the 64 pad bits are removed. The remaining 1 193 408 bits are the 116 rows of the 800GBASE-ER1 frame extended with a CRC32 every four rows. (see Figure 186–9).

### 186.2.4.5 CRC32 check and error marking

The CRC32 for each set of 4 rows is checked and removed. When errors are detected, the receiver determines which 257-bit blocks within the 4 rows protected by that CRC32 (41 120 bits) carry payload information and replaces only those 257-bit blocks with transcoded error control blocks (EBLOCK\_R), leaving any other 257-bit block unmodified. The details of how to determine which bits correspond to 257-bit blocks carrying payload information or overhead information from the 800GBASE-ER1 frame are beyond the scope of this standard.

The following counters shall be implemented:

FEC\_crc\_total\_blocks\_counter

The FEC\_crc\_total\_blocks\_counter is a 48-bit counter that counts once for each CRC32 block processed. This may be used together with the FEC\_crc\_errored\_blocks\_counter to calculate a CRC error ratio.

FEC\_crc\_errored\_blocks\_counter

The FEC\_crc\_corrected\_blocks\_counter is a 16-bit counter that counts once for each CRC32 block with errors. This may be used together with the FEC\_crc\_total\_blocks\_counter to calculate a CRC error ratio.

### 186.2.4.6 128-bit distribution

The set of 1 192 480 bits is already row-aligned with respect to the 800GBASE-ER1 frame. Groups of 128 bits are distributed to 8 FEC flows in a round-robin manner. The functions of FAM and OH field detection and removal and GMP demapping are performed on each of the 8 FEC flows.

### 186.2.4.7 FAM and OH fields detection and removal

The 800GBASE-ER1 frame floats within the ER1 FEC frame. The FAM field is the first 480 bits of each of the eight 800GBASE-ER1 tributary frames, and repeats every 1 320 980 bits. This field shall be aligned and locked using the 800GBASE-ER1 FEC sublayer FAM field lock state diagram (Figure 186–18).

Once FAM field lock has been acquired, the MFAS, status, and JC1–JC6 information is extracted from the OH field for use by the GMP demapper and for error signaling.

#### 186.2.4.7.1 MFAS alignment

The purpose of MFAS for the 800GBASE-ER1 receiver is to identify which 800GBASE-ER1 tributary frames contain the PT, MSI, and JC1–JC6 octets which are needed for the GMP demapping process and the AML octets which are needed for the optional alignment marker location feature. Alignment to the four-frame 800GBASE-ER1 multi-frame is achieved via the two LSBs of the MFAS. Alignment to the eight-frame 800GBASE-ER1 multi-frame is achieved via the three LSBs of the MFAS.

The overhead bytes over the four 800GBASE-ER1 tributary frame multi-frame are shown in Figure 186–6.

#### 186.2.4.7.2 Link status monitoring

The STAT octet present in every 800GBASE-ER1 tributary frame shall be monitored. It includes the remote PHY fault indication.

#### 186.2.4.7.3 Group management overhead, payload type and multiplex structure identifier

The CRC in columns 10 and 11 of the overhead frame (see Figure 186–6) is computed. If it is valid, the other overhead fields are extracted and processed as described in this subclause. If the CRC is not valid, these overheads are discarded and the prior received value of each field continues to be used.

The Group ID (GID) and Instance ID (IID) are extracted from the first frame of the eight-frame multi-frame. The MAP field is extracted from the eight-frame multi-frame. The expected values of these fields are as described in 186.2.3.5.3, 186.2.3.5.4, and 186.2.3.5.5, respectively.

The payload type (PT) is extracted from the fifth frame of the eight-frame multi-frame. The value indicates what signal is present in the payload area (800GE or test pattern). When test pattern generation is disabled by the FEC control variable tx\_test\_mode (set to 0), the value is expected to be 0x42. When test pattern generation is enabled by the FEC control variable tx\_test\_mode (set to 1), the value is expected to be 0x41.

The multiplex structure identifier (MSI) is extracted from the fifth frame of the eight-frame multi-frame. When test pattern generation is disabled by the FEC control variable tx\_test\_mode (set to 0), the value is expected to be 0x80. When test pattern generation is enabled by the FEC control variable tx\_test\_mode (set to 1), the value is expected to be 0x00.

When any of the GID, IID, MAP, PT, or MSI values do not match the expected value, the payload area of the frame is replaced with local fault ordered sets.

#### 186.2.4.7.4 GMP justification control overhead

The JC1-JC6 octets are found in the second, third, and fourth frames of the four-frame multi-frame. They are extracted and verified as described in Recommendation ITU-T G.709.1 Annex B, and used to control the operation of the GMP demapper (see 186.2.4.8).

#### 186.2.4.7.5 Alignment Marker Location control overhead (AML)

When the alignment marker location feature is supported as indicated by the FEC status variable FEC\_alignment\_marker\_location\_ability (set to 1) and enabled by the FEC control variable FEC\_alignment\_marker\_location\_enable (set to 1), the contents of the AML field are extracted from the third and fourth frames of the four-frame multi-frame. A non-zero value indicates the offset between the first 257-bit block that is demapped from the frame and the previous set of 800GBASE-R PCS alignment markers. Since the number of 257-bit blocks between alignment markers is known, the AML information can be used to determine when the next set of 800GBASE-R PCS alignment markers are to be inserted by the inverse RS-FEC function.

The value 0x000000 indicates that the transmit function did not encode any information about the location of alignment markers in this frame.

When alignment market location is not supported or not enabled, the AML field is ignored.

#### 186.2.4.7.6 Client status (CSTAT)

The client status information is found in byte 2 of the eighth frame of the eight-frame multi-frame. The information in bits 5-7 is extracted and used to populate the `am_sf<2:0>` bits of the alignment markers that are subsequently inserted according to the following rules:

- The value in `CSTAT<6>` is replicated to `am_sf<0>` of the two flows
- The value in `CSTAT<7>` is replicated to `am_sf<2>` of the two flows
- The value in `CSTAT<8>` is logically ORed with `FEC_degraded_SER` and replicated to `am_sf<1>` of the two flows

The information in bits 1-3 is extracted and used to determine the `SIGNAL_OK` parameter communicated via the `FEC:IS_SIGNAL.indication` primitive as described in Table 186–2.

#### 186.2.4.7.7 Other overhead

The other overheads (GID, IID, MAP) do not need to be processed. They contain fixed values and may be ignored by the receiver.

#### 186.2.4.8 GMP demapping

The GMP demapper shall decode the JC1-JC6 octets according to the procedures described in Recommendation ITU-T G.709.1 Annex B, recover the parameters  $C_m(t)$  and  $\Sigma C_{nD}(t)$ , and use them to recover the 257-bit data blocks that were inserted into the frame by the GMP mapper and the signal stream rate.

If the payload type or multiplex structure identification overhead (see 186.2.4.7.3) did not match the expected value, the 257-bit data blocks extracted from the GMP frame are replaced with LF ordered sets.

In addition to being protected by the CRC32, the JC1-JC2 field information is also protected by limits on how the JC1-JC2 fields might change in successive multi-frames (see Table 186–3) and the coding technique for indicating these changes, which combine with the CRC8 in JC3 and the CRC4 in JC6 to provide error correction capability for bit and burst errors impacting JC1-JC6.

#### 186.2.4.9 257-bit block collection

The 257-bit blocks from each of the eight FEC flows are interleaved, in order starting with FEC flow 0 and ending with FEC flow 7, to form a single stream of 257-bit blocks.

#### 186.2.4.10 Inverse RS-FEC Receive

The inverse RS-FEC Receive function distributes the stream of 257-bit blocks to two PCS flows and adds RS(544,514) FEC and alignment markers used by the 800GBASE-R PCS.

##### 186.2.4.10.1 Flow distribution

The stream of 257-bit blocks is distributed to two PCS flows (0 and 1) in a round-robin manner, with blocks from FEC flows 0, 2, 4, and 6 distributed to PCS flow 0, and blocks from FEC flows 1, 3, 5, and 7 distributed to PCS flow 1.

##### 186.2.4.10.2 Scrambler

The Scrambler function shall be as specified in 172.2.4.5.

### 186.2.4.10.3 Alignment marker mapping and insertion

The Alignment marker mapping and insertion function shall be as specified in 172.2.4.6, with the following exceptions:

- The bits `tx_am_sf<2:0>` are assigned as described in 186.2.4.7.6.
- The location of alignment markers is determined by counting 257-bit blocks between alignment markers. If the alignment marker location feature is supported (`FEC_alignment_marker_location_ability` is set to 1) and is enabled (`FEC_alignment_marker_location_enable` is set to 1), the location of the alignment markers is synchronized to information received in the AML field. The AMs are inserted at their original position, matching the exact location from which they were removed by the far-end transmit function, as indicated by the received AML value. When an unexpected AML value is received, the previous position of the AM is maintained until 8 consecutive unexpected AML values are received, after which the AM position is updated to the new position indicated by the received AML

### 186.2.4.10.4 Pre-FEC distribution

The Pre-FEC distribution function shall be as specified in 172.2.4.7.

### 186.2.4.10.5 Reed-Solomon encoder

The Reed-Solomon encoder function shall be as specified in 172.2.4.8.

### 186.2.4.10.6 Symbol distribution

The Symbol distribution function shall be as specified in 172.2.4.9.

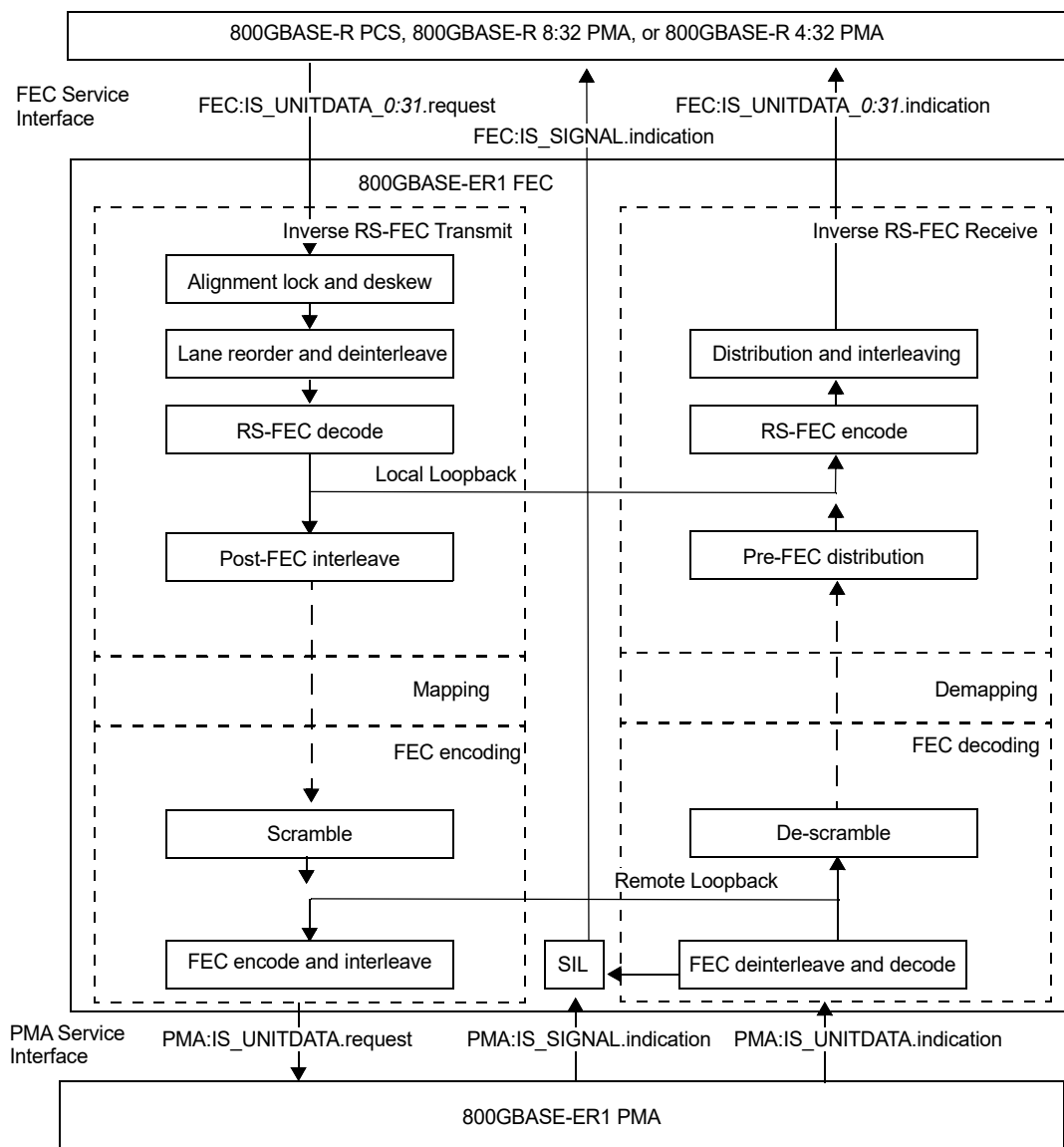
### 186.2.4.11 Test pattern checker

The FEC sublayer shall include eight PRBS31 test pattern checkers, one for each FEC flow presented by the GMP demapper. The group of eight PRBS31 test pattern checkers are enabled by the FEC control variable `rx_test_mode` (set to 1).

For the test configuration illustrated in Figure 174A–2, each PRBS31 test pattern checker instance shall include the block error detection and 17 related counters, as defined in Annex 174A.9.

### 186.2.5 Loopbacks

Loopback functions are optional. If implemented, they shall be as described in this subclause. The locations of the loopbacks with respect to the ER1 FEC functional block diagram (see Figure 186–2) are shown in Figure 186–12. The dashed lines in the transmit and receive functions represent functions from Figure 186–2 that are not shown because they are not relevant to the loopback locations



**Figure 186-12—800GBASE-ER1 FEC loopbacks**

### 186.2.5.1 Local loopback

The ER1 FEC sublayer may provide a local loopback. The function involves recovering the PCS lanes, decoding the RS FEC (186.2.3.1.3), and looping back each input PCS lane to the corresponding output PCS lane before the RS FEC encoder (186.2.4.10.5). During local loopback, the ER1 FEC sublayer transmit function performs the normal post-FEC interleave and subsequent functions toward the PMA service interface.

Ability to perform this function is indicated by the IFEC\_local\_loopback\_ability status variable. A device is placed in local loopback mode by setting the IFEC\_local\_loopback\_enable control variable to one, and removed from local loopback when this variable is set to zero.

### 186.2.5.2 Remote loopback

The ER1 FEC sublayer may provide a remote loopback. The function involves deinterleaving and decoding the ER1 FEC (186.2.4.1) and looping back each flow to the corresponding input of the FEC encoding and interleaving function (186.2.3.10). During remote loopback, the ER1 FEC sublayer receive function performs the normal descramble and subsequent functions toward the FEC service interface.

Ability to perform this function is indicated by the FEC\_remote\_loopback\_ability status variable. A device is placed in remote loopback mode by setting the FEC\_remote\_loopback\_enable control variable to one, and removed from remote loopback when this variable is set to zero.

### 186.2.6 Autonomous path startup (APSU) functions

The ER1 FEC shall provide the ready-to-send (RTS) function using the RTS without squelch state diagram (see 178B.11) as specified in Annex 178B.

The RTS status is signaled via the MNT bits using the mapping defined in 186.2.2. The variable c\_rts\_status is set as defined for FEC:IS\_SIGNAL.indication(SIGNAL\_OK) in Table 186–2.

The transmit function transmits ER1 FEC frames regardless of the value of c\_tx\_mode. When c\_tx\_mode = TRAINING, NOT-RTS, RTS, or FILL, the payload of the ER1 FEC frame is filled with the scrambled idle test pattern defined in 186.2.3.12.

The variable c\_isl\_ready is set to true when the FEC sublayer has aligned to the ER1 FEC frame and the ER1 tributary frame, and c\_local\_rx\_ready is true. Note that this requires that the PMA be aligned to the DSP frame. The variable c\_isl\_ready is set to false if the FEC sublayer has not aligned to the ER1 FEC frame and ER1 tributary frame, or if c\_local\_rx\_ready is false.

The variable c\_remote\_rts is set to true if the received SIGNAL\_OK status in the MNT bits from the peer interface indicate a status of READY or OK (see Table 186–2), and is set to false otherwise.

## 186.3 Physical Medium Attachment (PMA) sublayer, type 800GBASE-ER1

### 186.3.1 Overview

#### 186.3.1.1 Scope

This subclause specifies the Physical Medium Attachment (PMA) sublayer for the 800GBASE-ER1 and 800GBASE-ER1-20 PHYs. The 800GBASE-ER1 PMA allows the 800GBASE-ER1 FEC sublayer (see 186.2) to connect in a media independent way to the 800GBASE-ER1 or 800GBASE-ER1-20 PMD (specified in Clause 187).

The interfaces of the 800GBASE-ER1 PMA are defined in an abstract manner and do not imply any particular implementation.

#### 186.3.1.2 Position of the 800GBASE-ER1 PMA in the 800GBASE-ER1 sublayers

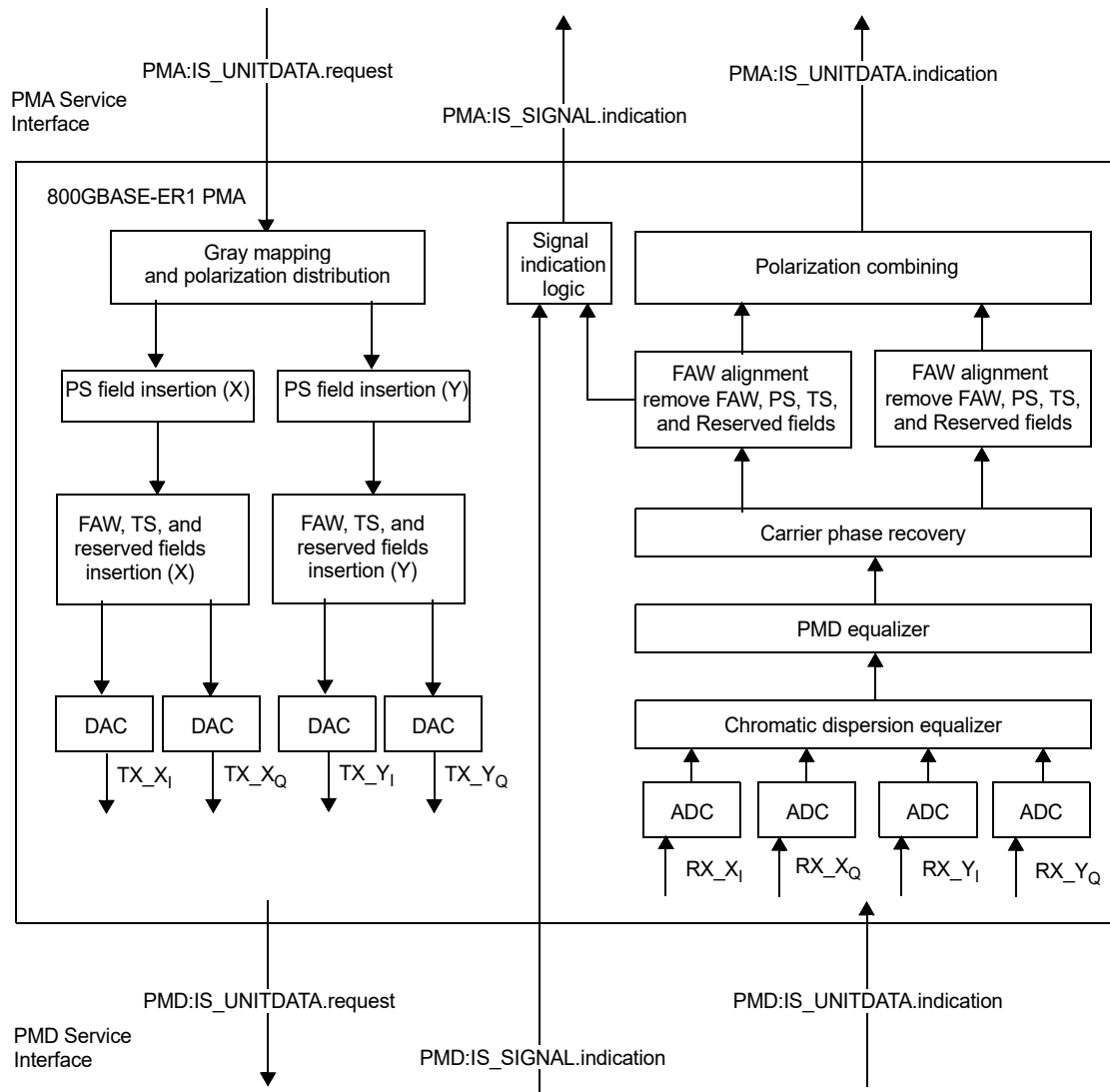
Figure 186–1 shows the relationship of the 800GBASE-ER1 PMA sublayer (shown shaded) with other sublayers to the ISO Open System Interconnection (OSI) reference model.

### 186.3.1.3 Summary of functions

The purpose of the 800GBASE-ER1 PMA is to adapt the 800GBASE-ER1 FEC sublayer digital signals to and from the four analog signals comprising each DP-16QAM symbol, and to recover timing information from the received signal. The following is a summary of the principal functions of the 800GBASE-ER1 PMA in both the transmit and receive directions:

- Gray mapping and distribution of symbols for transmission to the two polarizations, X and Y.
- Insertion of the pilot sequence (PS) field into the stream of symbols on each polarization.
- Insertion of the frame alignment word (FAW), training sequence (TS), and reserved fields into the stream of symbols on each polarization.
- Encoding and digital-to-analog conversion of the in-phase and quadrature components of the X and Y symbol streams.
- Sampling of the incoming signals from the 800GBASE-ER1 PMD coherent receiver.
- Signal processing of the incoming samples including: equalization of the incoming samples for the effects of chromatic dispersion and other fixed impairments caused by the link and by the receive portion of the 800GBASE-ER1 PMD; equalization of the incoming samples for dynamic impairments including both state of polarization and polarization mode dispersion; carrier phase and frequency recovery; and symbol timing recovery and retiming of the samples to the signaling rate.
- Synchronization to the FAW followed by: I-Q offset evaluation and compensation; and polarization offset evaluation and compensation.
- Removal of the PS field from the received stream of symbols on each polarization.
- Removal of the FAW, TS, and reserved fields from the received stream of symbols on each polarization.
- Polarization combining.
- Providing signal status information in the receive direction.

A functional diagram of the 800GBASE-ER1 PMA sublayer is shown in Figure 186–13. PMA implementations may differ from the structure shown in the functional diagram as long as the observable behavior of the 800GBASE-ER1 PHY is compliant.



**Figure 186-13—800GBASE-ER1 PMA functional block diagram**

### 186.3.2 800GBASE-ER1 PMA service interface

The 800GBASE-ER1 PMA service interface is an instance of the inter-sublayer service defined in 169.3.

The PMA service interface uses the following primitives to exchange codewords and status information:

```
PMA:IS_UNITDATA.request(tx_codeword)
PMA:IS_UNITDATA.indication(rx_codeword)
PMA:IS_SIGNAL.indication(SIGNAL_OK)
```

In the transmit direction, the PMA receives a 1 376 256-bit ER1 FEC codeword (see 186.2.3.10) from the 800GBASE-ER1 FEC sublayer as PMA:IS\_UNITDATA.request. In the receive direction, it conveys the best estimate of the in-phase and quadrature-phase of the X and Y polarization components of the 173 372 DP-16QAM symbols composing an ER1 FEC codeword in  $1\,376\,256 \times m$  bits as follows, where  $m$  is the

implementation-dependent resolution of the DP-16QAM symbol in bits and  $k$  is the DP-16QAM index with a range of 0 to 172 032.

The DP-16AM symbols are mapped to rx\_codeword as follows:

rx\_codeword $\langle (k \times 4 + 1) \times m - 1 : k \times 4 \times m \rangle$  are the in-phase component of the X polarization (XI)  
rx\_codeword $\langle (k \times 4 + 2) \times m - 1 : (k \times 4 + 1) \times m \rangle$  are the quadrature component of the X polarization (XQ)  
rx\_codeword $\langle (k \times 4 + 3) \times m - 1 : (k \times 4 + 2) \times m \rangle$  are the in-phase component of the Y polarization (YI)  
rx\_codeword $\langle (k \times 4 + 4) \times m - 1 : (k \times 4 + 3) \times m \rangle$  are the quadrature component of the X polarization (YQ)

The 800GBASE-ER1 PMA generates a PMA:IS\_UNITDATA.indicate message for each ER1 FEC codeword accumulated by the polarization combining and symbol deinterleaving function (see 186.3.4.4).

The PMA provides signal status information to the 800GBASE-ER1 FEC sublayer using the PMA:IS\_SIGNAL.indication primitive. The SIGNAL\_OK parameter takes one of two values:

- OK The 800GBASE-ER1 PMA receive function is not detecting a fault as defined in 186.3.4.5.
- FAIL The 800GBASE-ER1 PMA receive function is detecting a fault as defined in 186.3.4.5.

The 800GBASE-ER1 PMA generates a PMA:IS\_SIGNAL.indication message whenever there is change in the value of the SIGNAL\_OK parameter (see 186.3.4.5). The effect of the SIGNAL\_OK parameter when received by the 800GBASE-ER1 FEC sublayer is defined in 186.2.2.

### 186.3.3 800GBASE-ER1 PMA transmit function

The input of the 800GBASE-ER1 PMA transmit function receives 1 376 256-bit ER1 FEC codewords at 1/172 032 the DP-16QAM signaling rate from the 800GBASE-ER1 FEC sublayer. The output signals between the 800GBASE-ER1 PMA transmit function and 800GBASE-ER1 or 800GBASE-ER1-20 PMD carry analog signals representing the components of DP-16QAM symbols (namely  $X_I$ ,  $X_Q$ ,  $Y_I$ , and  $Y_Q$ ).

#### 186.3.3.1 Gray mapping and polarization distribution

The 800GBASE-ER1 PMA transmit function shall map sets of eight bits to DP-16QAM codewords, with a pair of bits mapped to each of the four components of the DP-16QAM symbol using Gray coding. Each ER1 FEC codeword  $[c_0, c_1, \dots, c_{1376255}]$  passed across the PMA service interface from the ER1 FEC encoder in the tx\_codeword parameter of the PMA:IS\_UNITDATA.request primitive is mapped to 172 032 DP-16QAM symbols  $[S_0, S_1, \dots, S_{172031}]$  where, for  $i = 0$  to 172 031:

- $(c_{8i}, c_{8i+2})$  maps to the in-phase (I) component of the X-polarization of  $S_i$
- $(c_{8i+4}, c_{8i+6})$  maps to the quadrature-phase (Q) component of the X-polarization of  $S_i$
- $(c_{8i+1}, c_{8i+3})$  maps to the in-phase (I) component of the Y-polarization of  $S_i$
- $(c_{8i+5}, c_{8i+7})$  maps to the quadrature-phase (Q) component of the Y-polarization of  $S_i$

The complete mapping between the bits of ER1 FEC codewords and Gray-coded DP-16QAM symbols is detailed in Table 186–4

**Table 186–4—In-phase (I) and quadrature-phase (Q) symbol values**

| $X: (c_{8i}, c_{8i+2}, c_{8i+4}, c_{8i+6})$<br>$Y: (c_{8i+1}, c_{8i+3}, c_{8i+5}, c_{8i+7})$ | $X_I$<br>$Y_I$ | $X_Q$<br>$Y_Q$ |
|----------------------------------------------------------------------------------------------|----------------|----------------|
| (0,0,0,0)                                                                                    | –3             | –3             |
| (0,0,0,1)                                                                                    | –3             | –1             |
| (0,0,1,0)                                                                                    | –3             | 3              |
| (0,0,1,1)                                                                                    | –3             | 1              |
| (0,1,0,0)                                                                                    | –1             | –3             |
| (0,1,0,1)                                                                                    | –1             | –1             |
| (0,1,1,0)                                                                                    | –1             | 3              |
| (0,1,1,1)                                                                                    | –1             | 1              |
| (1,0,0,0)                                                                                    | 3              | –3             |
| (1,0,0,1)                                                                                    | 3              | –1             |
| (1,0,1,0)                                                                                    | 3              | 3              |
| (1,0,1,1)                                                                                    | 3              | 1              |
| (1,1,0,0)                                                                                    | 1              | –3             |
| (1,1,0,1)                                                                                    | 1              | –1             |
| (1,1,1,0)                                                                                    | 1              | 3              |
| (1,1,1,1)                                                                                    | 1              | 1              |

The 175 104 DP-16QAM symbols for the X and Y polarizations are distributed to two streams of 175 104 16QAM symbols for further processing.

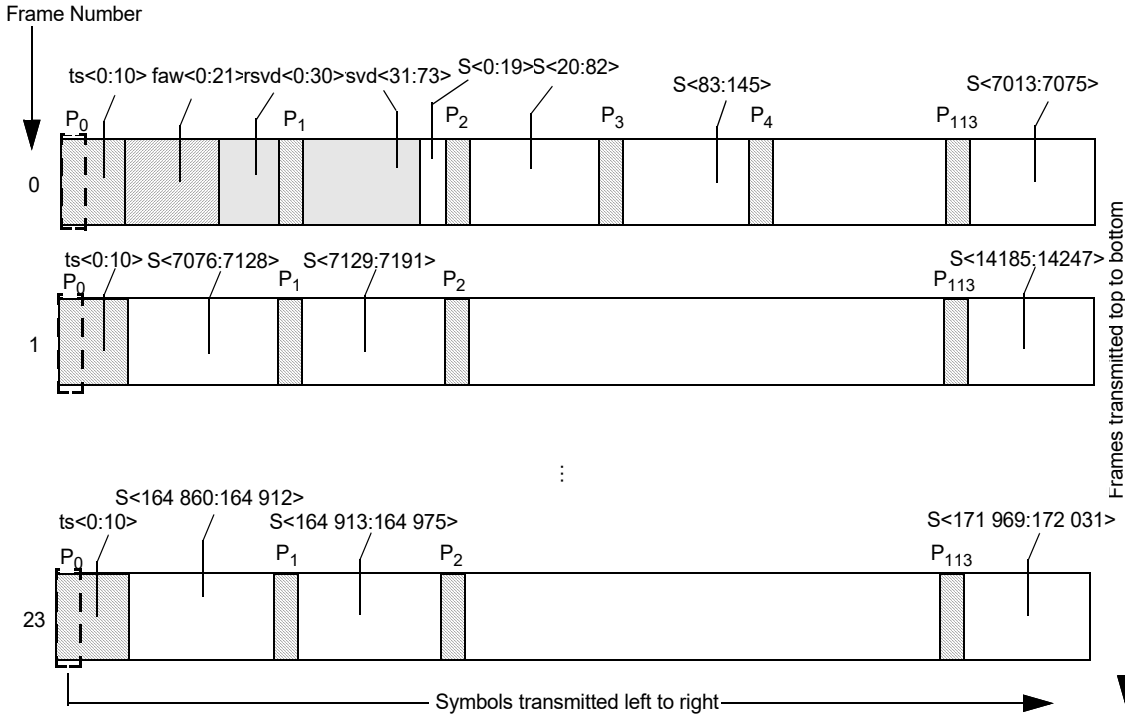
### 186.3.3.2 800GBASE-ER1 PMA frame and multi-frame assembly

The stream of 16QAM symbols for each polarization is assembled into the 800GBASE-ER1 PMA frame and multi-frame. The frame structure consists of 7296 symbols that are organized into 114 rows of 64 symbols each. The multi-frame consists of 24 frames, for a total of 175 104 symbols per multi-frame. The details of the frame and multi-frame structures are defined in clause 11 and 12.1 of [ITU-T G.709.6], respectively, with the parameter  $k = 8$ .

Each 800GBASE-ER1 PMA frame contains 114 rows of 64 symbols. The first symbol of each row is a pilot symbol [P0, P1, ..., P113]. Each 800GBASE-ER1 PMA frame begins with an 11-symbol TS field carrying a training sequence (ts<0:10>). The first symbol of the training sequence is the pilot symbol P0.

The first 800GBASE-ER1 PMA frame in the multi-frame includes a 22-symbol multi-frame alignment field (faw<0:21>), a 74-symbol reserved field (rsvd<0:73>), and 7076 payload symbols (S<0:7075>). All other frames in the multi-frame have 7172 payload symbols (63 in each row, following the pilot symbol).

Each 800GBASE-ER1 PMA multi-frame shall be serialized into a stream of 16QAM symbols for transmission. Relative to Figure 186–14, the 800GBASE-ER1 PMA frames shall be transmitted from top to bottom, and the symbols of each 800GBASE-ER1 PMA frame shall be transmitted from left to right. The assembly of symbols into 800GBASE-ER1 PMA frames and multi-frames is continuous.



**Figure 186–14—800GBASE-ER1 PMA Multi-frame and frame formats**

### 186.3.3.3 Pilot sequence

The pilot sequence is a fixed PRBS10 sequence mapped to 16QAM symbols with different seed values for X and Y polarizations, as described in Annex A.1.2 of [ITU-T G.709.6]. The generator for each pilot sequence is shown in Figure 186–15, with the polynomial and seed values also indicated in Table 186–5. The seeds are selected so that the pilot and training sequences, when combined, are DC balanced. The seeds are also selected so that the first symbol in the training sequence is also the first symbol in the pilot sequence.

The generator is initialized using the seed at the start of every 800GBASE-ER1 PMA frame, so that the same 114 symbols,  $[P_0, \dots, P_{113}]$  are inserted into every 800GBASE-ER1 PMA frame PS field of the same polarization. For each polarization X and Y, the generator produces 228 bits PRBS[227:0] that are mapped to 114 16QAM symbols,  $[P_0, \dots, P_{113}]$  where for  $i = 0$  to 113:

- PRBS[2*i*] maps to the in-phase (I) component of the 16QAM symbol  $[P_i]$  for the respective polarization
- PRBS[2*i*+1] maps to the quadrature-phase (Q) component of the 16QAM symbol  $[P_i]$  for the respective polarization

and where,

- 0 maps to  $-3$  for the respective 16QAM symbol component
- 1 maps to  $+3$  for the respective 16QAM symbol component

The complete pilot sequence is shown in Table 186–6.

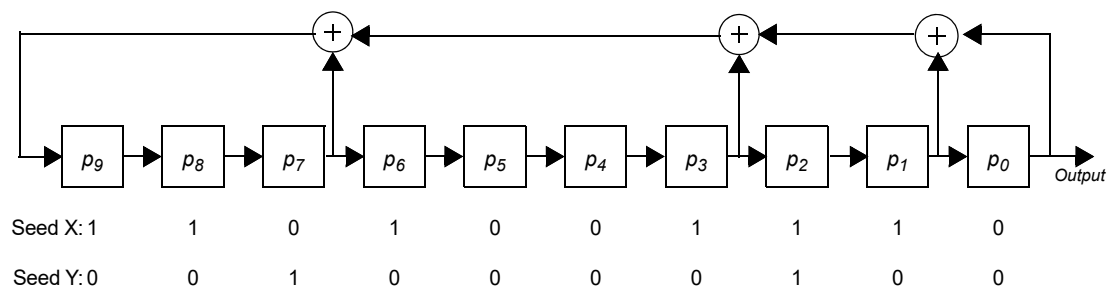


Figure 186–15—Pilot seed and sequence

Table 186–5—Pilot sequence generator polynomial and seed values

| Generator polynomial         | Seed for X-polarization | Seed for Y-polarization |
|------------------------------|-------------------------|-------------------------|
| $x^{10} + x^7 + x^3 + x + 1$ | 0x34E                   | 0x084                   |

Table 186–6—Pilot sequence

| Index | X <sub>I</sub> | X <sub>Q</sub> | Y <sub>I</sub> | Y <sub>Q</sub> | Index | X <sub>I</sub> | X <sub>Q</sub> | Y <sub>I</sub> | Y <sub>Q</sub> | Index | X <sub>I</sub> | X <sub>Q</sub> | Y <sub>I</sub> | Y <sub>Q</sub> | Index | X <sub>I</sub> | X <sub>Q</sub> | Y <sub>I</sub> | Y <sub>Q</sub> |
|-------|----------------|----------------|----------------|----------------|-------|----------------|----------------|----------------|----------------|-------|----------------|----------------|----------------|----------------|-------|----------------|----------------|----------------|----------------|
| 0     | –3             | 3              | –3             | –3             | 29    | 3              | –3             | –3             | 3              | 58    | 3              | –3             | –3             | 3              | 87    | –3             | 3              | –3             | –3             |
| 1     | 3              | 3              | 3              | –3             | 30    | 3              | 3              | 3              | 3              | 59    | 3              | –3             | –3             | 3              | 88    | 3              | 3              | 3              | 3              |
| 2     | –3             | –3             | –3             | –3             | 31    | 3              | –3             | 3              | 3              | 60    | –3             | –3             | –3             | –3             | 89    | 3              | –3             | 3              | 3              |
| 3     | 3              | –3             | –3             | 3              | 32    | –3             | 3              | –3             | –3             | 61    | 3              | 3              | –3             | –3             | 90    | 3              | –3             | 3              | –3             |
| 4     | 3              | 3              | –3             | –3             | 33    | –3             | –3             | –3             | 3              | 62    | 3              | –3             | 3              | 3              | 91    | –3             | –3             | 3              | 3              |
| 5     | –3             | 3              | 3              | 3              | 34    | 3              | –3             | –3             | 3              | 63    | –3             | –3             | –3             | –3             | 92    | –3             | 3              | 3              | 3              |
| 6     | 3              | –3             | 3              | 3              | 35    | –3             | 3              | –3             | –3             | 64    | 3              | 3              | –3             | 3              | 93    | –3             | 3              | –3             | –3             |
| 7     | 3              | 3              | –3             | 3              | 36    | –3             | –3             | 3              | 3              | 65    | 3              | 3              | –3             | 3              | 94    | –3             | 3              | 3              | 3              |
| 8     | –3             | –3             | –3             | –3             | 37    | –3             | –3             | 3              | 3              | 66    | 3              | 3              | –3             | 3              | 95    | –3             | –3             | –3             | 3              |
| 9     | –3             | –3             | –3             | –3             | 38    | –3             | 3              | 3              | –3             | 67    | –3             | –3             | 3              | 3              | 96    | –3             | –3             | –3             | –3             |
| 10    | 3              | 3              | 3              | –3             | 39    | –3             | –3             | –3             | 3              | 68    | –3             | –3             | –3             | 3              | 97    | –3             | 3              | 3              | –3             |
| 11    | –3             | –3             | 3              | –3             | 40    | –3             | 3              | 3              | –3             | 69    | 3              | 3              | 3              | 3              | 98    | –3             | 3              | 3              | –3             |
| 12    | 3              | 3              | 3              | –3             | 41    | –3             | –3             | –3             | –3             | 70    | 3              | 3              | 3              | 3              | 99    | –3             | 3              | 3              | –3             |
| 13    | –3             | –3             | –3             | –3             | 42    | –3             | –3             | –3             | –3             | 71    | 3              | –3             | 3              | 3              | 100   | 3              | –3             | 3              | 3              |
| 14    | –3             | 3              | –3             | –3             | 43    | 3              | 3              | –3             | 3              | 72    | 3              | –3             | –3             | 3              | 101   | –3             | 3              | –3             | 3              |
| 15    | –3             | –3             | 3              | –3             | 44    | 3              | 3              | 3              | –3             | 73    | 3              | 3              | 3              | 3              | 102   | –3             | 3              | –3             | –3             |
| 16    | –3             | 3              | 3              | –3             | 45    | –3             | –3             | –3             | 3              | 74    | 3              | –3             | –3             | –3             | 103   | 3              | –3             | –3             | –3             |
| 17    | –3             | 3              | 3              | 3              | 46    | 3              | 3              | –3             | –3             | 75    | 3              | –3             | –3             | –3             | 104   | –3             | 3              | –3             | 3              |

**Table 186–6—Pilot sequence**

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|----|----|----|----|
| 18 | −3 | −3 | −3 | −3 | 47 | 3  | 3  | −3 | 3  | 76 | 3  | 3  | 3  | −3 | 105 | −3 | −3 | 3  | 3  |
| 19 | −3 | 3  | 3  | −3 | 48 | −3 | 3  | 3  | −3 | 77 | −3 | 3  | −3 | 3  | 106 | 3  | −3 | −3 | −3 |
| 20 | 3  | −3 | 3  | 3  | 49 | 3  | −3 | 3  | 3  | 78 | 3  | 3  | −3 | 3  | 107 | 3  | 3  | 3  | −3 |
| 21 | 3  | −3 | −3 | −3 | 50 | −3 | 3  | 3  | −3 | 79 | −3 | 3  | 3  | 3  | 108 | −3 | 3  | 3  | −3 |
| 22 | 3  | −3 | 3  | −3 | 51 | 3  | −3 | −3 | −3 | 80 | 3  | 3  | 3  | −3 | 109 | −3 | −3 | −3 | 3  |
| 23 | 3  | −3 | −3 | 3  | 52 | −3 | 3  | 3  | 3  | 81 | −3 | 3  | 3  | 3  | 110 | 3  | −3 | −3 | −3 |
| 24 | 3  | −3 | −3 | 3  | 53 | 3  | −3 | −3 | 3  | 82 | 3  | −3 | −3 | 3  | 111 | −3 | −3 | 3  | −3 |
| 25 | 3  | 3  | 3  | −3 | 54 | 3  | −3 | 3  | −3 | 83 | −3 | 3  | 3  | 3  | 112 | 3  | 3  | 3  | −3 |
| 26 | 3  | 3  | 3  | −3 | 55 | 3  | 3  | −3 | −3 | 84 | −3 | −3 | 3  | −3 | 113 | 3  | 3  | −3 | 3  |
| 27 | −3 | 3  | −3 | 3  | 56 | −3 | −3 | 3  | 3  | 85 | −3 | −3 | 3  | −3 |     |    |    |    |    |
| 28 | −3 | −3 | −3 | −3 | 57 | −3 | 3  | 3  | 3  | 86 | 3  | −3 | 3  | 3  |     |    |    |    |    |

#### 186.3.3.4 Training sequence

An 11-symbol training sequence as defined in Annex A.1.1 of [ITU-T G.709.6] is added at the beginning (transmitted first) of each 800GBASE-ER1 PMA multi-frame. The training sequences for X and Y polarizations are different in order to aid in identification by receivers. The first symbol of the training sequence is processed as part of the pilot sequence. It is made up of the outer four points of the 16QAM constellation symbol values and are listed in Table 186–7.

**Table 186–7—Training sequences**

| Index | X <sub>I</sub> | X <sub>Q</sub> | Y <sub>I</sub> | Y <sub>Q</sub> |
|-------|----------------|----------------|----------------|----------------|
| 0     | −3             | 3              | −3             | −3             |
| 1     | 3              | 3              | −3             | −3             |
| 2     | −3             | 3              | 3              | −3             |
| 3     | 3              | 3              | −3             | 3              |
| 4     | −3             | −3             | −3             | 3              |
| 5     | 3              | 3              | 3              | 3              |
| 6     | −3             | −3             | −3             | −3             |
| 7     | −3             | −3             | −3             | 3              |
| 8     | 3              | 3              | 3              | −3             |
| 9     | 3              | −3             | 3              | 3              |
| 10    | 3              | −3             | 3              | −3             |

### 186.3.3.5 Multi-frame alignment word (FAW) and Reserved

The 22-symbol multi-frame alignment word defined in Annex A.1.3 of [ITU-T G.709.6] is inserted in the first frame of each multi-frame. The values for each polarization are listed in terms of constellation points in Table 186–8. It is made up of the outer four points of the 16QAM constellation symbol values and designed for DC balance. The receive function aligns to the FAW field in order to identify the 800GBASE-ER1 PMA multi-frame boundaries and for skew and skew variation removal between polarizations.

Following the FAW, a reserved field of 74 symbols is inserted. The value of the reserved symbols should be randomized, but is not otherwise specified.

**Table 186–8—Multi-frame alignment word symbols**

| Index | $X_I$ | $X_Q$ | $Y_I$ | $Y_Q$ |
|-------|-------|-------|-------|-------|
| 0     | 3     | –3    | 3     | 3     |
| 1     | 3     | 3     | –3    | 3     |
| 2     | 3     | 3     | –3    | –3    |
| 3     | 3     | 3     | –3    | 3     |
| 4     | 3     | –3    | 3     | –3    |
| 5     | 3     | –3    | 3     | 3     |
| 6     | –3    | –3    | 3     | –3    |
| 7     | 3     | 3     | 3     | –3    |
| 8     | –3    | –3    | –3    | –3    |
| 9     | –3    | 3     | 3     | –3    |
| 10    | –3    | 3     | 3     | 3     |
| 11    | 3     | –3    | –3    | 3     |
| 12    | –3    | –3    | –3    | 3     |
| 13    | –3    | –3    | 3     | 3     |
| 14    | –3    | 3     | –3    | –3    |
| 15    | 3     | 3     | 3     | 3     |
| 16    | –3    | –3    | –3    | –3    |
| 17    | 3     | –3    | –3    | 3     |
| 18    | –3    | 3     | 3     | –3    |
| 19    | 3     | 3     | –3    | –3    |
| 20    | –3    | –3    | 3     | –3    |
| 21    | –3    | 3     | –3    | 3     |

### 186.3.3.6 Analog signal generation

The two streams of symbols, one for each polarization, shall be converted to four analog signals, two for each polarization:  $X_I$ ,  $X_Q$ ,  $Y_I$ , and  $Y_Q$ . The analog signals are passed to the 800GBASE-ER1 or 800GBASE-ER1-20 PMD sublayer using the tx\_signal\_xi, tx\_signal\_xq, tx\_signal\_yi, tx\_signal\_yq parameters of the PMD:IS\_UNITDATA.request primitive.

### 186.3.3.7 Transmit bit ordering and distribution

The transmit bit ordering and distribution for the 800GBASE-ER1 PMA are illustrated in Figure 186–16.

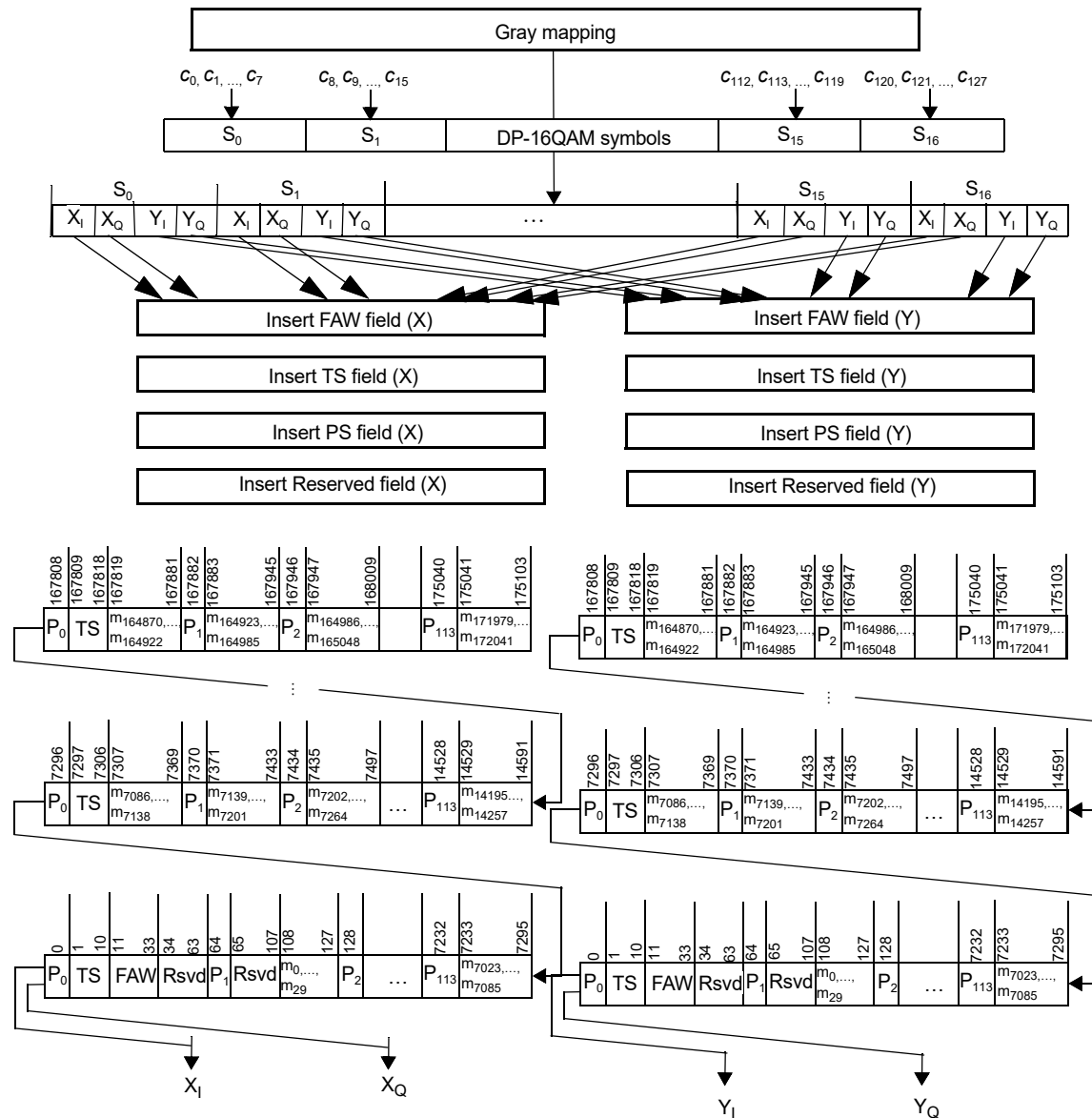


Figure 186–16—800GBASE-ER1 PMA transmit bit ordering and distribution

#### 186.3.4 800GBASE-ER1 PMA receive function

The input of the 800GBASE-ER1 PMA receive function receives analog signals representing the components of DP-16QAM symbols (namely  $X_I$ ,  $X_Q$ ,  $Y_I$ , and  $Y_Q$ ) from the 800GBASE-ER1 or 800GBASE-ER1-20 PMD.

The output of the 800GBASE-ER1 PMA receive function provides  $1\ 376\ 256 \times m$  bits representing the ER1 FEC decoder input at  $1/172\ 032$  the DP-16QAM signaling rate to the 800GBASE-ER1 FEC.

##### 186.3.4.1 Receive signal sampling

Four analog signals  $RX\_X_I$ ,  $RX\_X_Q$ ,  $RX\_Y_I$ , and  $RX\_Y_Q$  are received from 800GBASE-ER1 or 800GBASE-ER1-20 PMD using the `rx_signal_xi`, `rx_signal_xq`, `rx_signal_yi`, `rx_signal_yq` parameters of the `PMD:IS_UNITDATA.indication` primitive. Each parameter of the `PMD:IS_UNITDATA.indication` primitive shall be sampled at a sampling rate equal to or larger than the 800GBASE-ER1 signaling rate. The resulting samples are forwarded to the receive signal processing block.

The details of the sampling, including any quantization and the chosen sampling rate are implementation specific.

##### 186.3.4.2 Receive signal processing

The four incoming streams of samples shall be processed and converted into two streams of 16QAM symbols representing the X and Y polarizations of the DP-16QAM symbols. The 16QAM symbols should be sampled with more than two bits per dimension, in order to enable the ER1 FEC decoder to correct errors and recover the bits from the symbols based on the mapping in Table 186–4.

Functions within the signal processing block include, but are not limited to:

- Retiming and resampling to one sample per symbol after applying an anti-aliasing filter with a –3 dB bandwidth of 0.5 times the signaling rate to the incoming samples.
- Equalization of signal distortion caused by chromatic dispersion, group delay and polarization mode dispersion in the channel.
- Clock phase recovery.
- Frequency offset estimation and removal assuming a constant frequency offset over a given block size  $N$  which is implementation specific.
- Carrier phase recovery.
- Identification of the X and Y polarizations and I and Q component of each polarization.
- I-Q offset evaluation and compensation.

The details of the receive signal processing are implementation specific. A PMD in combination with the PMA and ER1 FEC is required to meet the CRC error ratio specifications in 187.2.

##### 186.3.4.3 FAW, TS, Reserved, and PS fields removal

The 800GBASE-ER1 PMA receive path attains lock to the 22-symbol FAW field that is transmitted on each of the two transmission polarizations on the in-phase and quadrature-phase signals. The FAW field appears in the first frame of each 24-frame multi-frame as shown in Figure 186–14. After FAW fields are found on each of the receive polarization, the transmit X and Y polarization symbol streams are identified and deskewed. Refer to 187.6 for skew specifications.

After the X and Y polarization symbol streams are identified and aligned, the FAW, TS, Reserved, and PS fields shall be removed leaving the remaining 172 032 message symbols for further processing.

#### 186.3.4.4 Polarization combining and symbol deinterleaving

The message symbols from the X and Y polarization streams shall be combined to form ER1 FEC codewords comprising 1 376 256 symbols encoded as shown in Table 186–4 but at a higher resolution in order to aid the ER1 FEC error detection and correction process.

The 1 376 256 encoded DP-16QAM symbols are transferred to the 800GBASE-ER1 FEC sublayer as *m*-bit DP-16QAM symbols using PMA:IS\_UNITDATA.indication messages.

#### 186.3.4.5 Signal indication logic

The 800GBASE-ER1 PMA receive signal status is conveyed using the SIGNAL\_OK parameter via the PMA:IS\_SIGNAL.indication primitive (see 186.3.2).

SIGNAL\_OK is set to OK if all of the following conditions are met:

- the SIGNAL\_OK parameters in PMD:IS\_SIGNAL.indication primitive is OK
- data is being successfully processed by all 800GBASE-ER1 PMA receive signal processing functions

Otherwise, the SIGNAL\_OK parameter is set to the value FAIL.

### 186.4 State diagrams

This subclause is composed of state diagrams, including the associated definitions of constants, variables, functions, and counters.

#### 186.4.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

#### 186.4.2 State diagram definitions

##### 186.4.2.1 Constants

AM\_INTERVAL

The number of 257-bit blocks between alignment markers. For 800GBASE-R it is set to 327 664.

##### 186.4.2.2 Variables

aml\_value

A variable that holds the 24-bit value extracted from the AML field of the 800GBASE-ER1 tributary frame (see 186.2.3.5.10).

block\_rx

Boolean variable that is set to true when the next non-stuff 257b block is demapped by the GMP demapper function. It is set to false upon entering the RAML\_CNT\_0 or RAML\_CNT\_INC states in the 800GBASE-ER1 FEC sublayer alignment marker location state diagram (Figure 186–20).

current\_fam<*x*>

A variable that holds the alignment pattern corresponding to the current FAM overhead that is recognized on a given 800GBASE-ER1 FEC sublayer tributary FEC flow, where *x* = 0 to 7. It is compared to the variable first\_fam<*x*> to confirm the location of the FAM field has been detected.

|                                                                                                       |    |
|-------------------------------------------------------------------------------------------------------|----|
| current_mfas<x>                                                                                       | 1  |
| A variable that holds the MFAS pattern corresponding to the current MFAS overhead that is             | 2  |
| recognized on a given 800GBASE-ER1 FEC sublayer tributary FEC flow, where $x = 0$ to 7. It is         | 3  |
| compared to the variable first_mfas<x> to confirm multi-frame alignment.                              | 4  |
|                                                                                                       | 5  |
| current_pma_pss<y>                                                                                    | 6  |
| A variable that holds the 800GBASE-ER1 PMA polarization symbol stream identifier                      | 7  |
| corresponding to the current frame alignment word sequence that is recognized on a given              | 8  |
| polarization symbol stream of the 800GBASE-ER1 PMD service interface, where $y = 0$ to 1. It is       | 9  |
| compared to the variable first_pma_pss to confirm that the location of the frame alignment word       | 10 |
| sequence has been detected.                                                                           | 11 |
| Values:                                                                                               | 12 |
| See first_pma_pss.                                                                                    | 13 |
|                                                                                                       | 14 |
| fam_counter_done<x>                                                                                   | 15 |
| A Boolean variable that is set to true when fam_counter<x> has reached its terminal count, where      | 16 |
| $x = 0$ to 7. It is set to false when the counter is started by the 800GBASE-ER1 FEC sublayer FAM     | 17 |
| field lock state diagram (Figure 186–18).                                                             | 18 |
|                                                                                                       | 19 |
| fam_match<x>                                                                                          | 20 |
| A Boolean variable holds the output of the function FAM_COMPARE, where $x = 0$ to 7.                  | 21 |
|                                                                                                       | 22 |
| fam_slip_done<x>                                                                                      | 23 |
| A Boolean variable that indicates the next candidate 480-bit block position is available to be tested | 24 |
| to determine if it contains the frame alignment overhead for an 800GBASE-ER1 sublayer tributary       | 25 |
| FEC flow, where $x = 0$ to 7. It is set to true when the FAM_SLIP function completes and is set to    | 26 |
| false upon entering the GET_BLOCK state of the 800GBASE-ER1 FEC sublayer FAM field lock               | 27 |
| state diagram (Figure 186–18).                                                                        | 28 |
|                                                                                                       | 29 |
| fam_valid<x>                                                                                          | 30 |
| A Boolean variable that is set to true if the first 256 bits of the received 480-bit block on an      | 31 |
| 800GBASE-ER1 sublayer tributary FEC flow, where $x = 0$ to 7, are a valid 800GBASE-ER1 FEC            | 32 |
| tributary frame alignment mechanism sequence as described in 186.2.3.4.1, and set to false            | 33 |
| otherwise. The sequence is considered to be valid if all bits match the known bits of the pattern     | 34 |
| described in 186.2.3.4.1.                                                                             | 35 |
|                                                                                                       | 36 |
| fam_lock<x>                                                                                           | 37 |
| A Boolean variable that indicates the receiver has detected the location of the FAM field among the   | 38 |
| stream of 257-bit blocks on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where $x = 0$ to         | 39 |
| 7. The value of fam_lock<x> is set by the 800GBASE-ER1 FEC sublayer FAM field lock state              | 40 |
| diagram (Figure 186–18).                                                                              | 41 |
|                                                                                                       | 42 |
| faw_counter_done<y>                                                                                   | 43 |
| A Boolean variable that is set to true when faw_counter<y> has reached its terminal count, where      | 44 |
| $y = 0$ to 1. It is set to false when the counter is started by the 800GBASE-ER1 PMA FAW field        | 45 |
| lock state diagram (Figure 186–17).                                                                   | 46 |
|                                                                                                       | 47 |
| faw_lock<y>                                                                                           | 48 |
| A Boolean variable that indicates the receiver has detected the location of the FAW field for a       | 49 |
| given polarization symbol stream on the 800GBASE-ER1 PMD service interface, where $y = 0$ to 1.       | 50 |
| The value is set by the 800GBASE-R PMA FAW field lock state diagram (Figure 186–17).                  | 51 |
|                                                                                                       | 52 |
| faw_match<y>                                                                                          | 53 |
| A Boolean variable that holds the output of the function FAW_COMPARE for a received                   | 54 |
| polarization symbol stream, where $y = 0$ to 1.                                                       |    |
|                                                                                                       |    |
| faw_slip_done<y>                                                                                      |    |
| A Boolean variable that indicates the next candidate 22-symbol block position is available for        |    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| testing, where $y = 0$ to 1. It is set to true when the FAW_SLIP function completes and set to false upon entering the GET_BLOCK state of the 800GBASE-ER1 PMA FAW field lock state diagram (Figure 186–17).                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3  |
| faw_valid<y>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4  |
| A Boolean variable that is set to true if, after removing the PS, the candidate 22 16QAM symbol block from the received polarization symbol stream, where $y = 0$ to 1, is a valid frame alignment word sequence, and is set to false otherwise. The candidate 22 16QAM symbol block is compared to the frame alignment word sequences defined in Table 186–8, considering all permitted 800GBASE-ER1 PMD service interface physical signal mappings. The candidate 22 16QAM symbol block is considered to be a valid frame alignment word sequence if at least 18 symbols match the 22 known symbols of the frame alignment word sequence described in 186.3.3.5. | 5  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 7  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 8  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 9  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 11 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 12 |
| fec_all_locked                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 13 |
| A Boolean variable that is set to true when fam_lock<x> is true for all FEC flows and is set to false when fam_lock<x> is false for any x.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 14 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 15 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 16 |
| fec_all_mfas_locked                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 17 |
| A Boolean variable that is set to true when mfas_lock<x> is true for all FEC flows and is set to false when mfas_lock<x> is false for any x.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 18 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 19 |
| fec_mfas_restart_lock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 20 |
| A Boolean variable that is use to restart the MFAS field lock process when 5 consecutive MFAS values fail to match the expected value on a given 800GBASE-ER1 FEC sublayer tributary FEC flow. The value of fec_mfas_restart_lock is set by the 800GBASE-ER1 FEC sublayer multi-frame alignment lock state diagram (Figure 186–19).                                                                                                                                                                                                                                                                                                                                | 21 |
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| FEC_reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 26 |
| Boolean variable that is true when set by a management entity and is false otherwise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 27 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 28 |
| fec_restart_lock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 29 |
| A Boolean variable that is used to restart the FAM field lock process to when 5 consecutive FEC frame alignment mechanism patterns fail to match on a given 800GBASE-ER1 tributary FEC flow. The value of fec_restart_lock is set by the 800GBASE-ER1 FEC sublayer FAM lock field state diagram (Figure 186–18).                                                                                                                                                                                                                                                                                                                                                   | 30 |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 33 |
| fec_signal_ok                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 34 |
| A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the PMA:IS_SIGNAL.indication primitive. It is true if the value was OK and false if the value was FAIL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 35 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 36 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 37 |
| first_fam<x>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 38 |
| A variable that holds the alignment pattern corresponding to the first FAM overhead that is recognized on a given 800GBASE-ER1 FEC tributary FEC flow, per the 800GBASE-ER1 FEC sublayer FAM lock field state diagram (Figure 186–18), where $x = 0$ to 7. It is compared to the next received FEC frame alignment mechanism pattern that is tested.                                                                                                                                                                                                                                                                                                               | 39 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 40 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 41 |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 43 |
| first_mfas<x>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 44 |
| A variable that holds the MFAS pattern corresponding to the first MFAS overhead that is recognized on a given 800GBASE-ER1 FEC sublayer tributary FEC flow, where $x = 0$ to 7. It is compared to the next received MFAS pattern that is tested.                                                                                                                                                                                                                                                                                                                                                                                                                   | 45 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 46 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 47 |
| first_pma_pss<y>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 48 |
| A variable that holds the 800GBASE-ER1 PMA receive polarization symbol stream identifier corresponding to the first frame alignment word sequence that is recognized on a given physical signal of the 800GBASE-ER1 PMD service interface, per the 800GBASE-ER1 PMA FAW field lock state diagram (Figure 186–17), where $y = 0$ to 1. It is compared to the 800GBASE-ER1 PMA receive polarization symbol stream identifier corresponding to the next frame alignment word sequence that is tested.                                                                                                                                                                 | 49 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 50 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 51 |
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|                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Values:                                                                                                                                                                                                                                                                                                                                                                                                          | 1  |
| X: Value of the frame alignment word sequence for the given 800GBASE-ER1 PMA receive polarization symbol stream maps to Table 186–8 sequence $X_I$ , $X_Q$ .                                                                                                                                                                                                                                                     | 2  |
| Y: Value of the frame alignment word sequence for the given 800GBASE-ER1 PMA receive polarization symbol stream maps to Table 186–8 sequence $Y_I$ , $Y_Q$ .                                                                                                                                                                                                                                                     | 3  |
| frame_counter_done<x>                                                                                                                                                                                                                                                                                                                                                                                            | 4  |
| A Boolean variable that is set to true when frame_counter<x> has reached its terminal count, where $x = 0$ to 7. It is set to false when the counter is started by the 800GBASE-ER1 FEC sublayer multi-frame alignment state diagram (Figure 186–19).                                                                                                                                                            | 5  |
| mfas_lock<x>                                                                                                                                                                                                                                                                                                                                                                                                     | 6  |
| A Boolean variable that indicates the receiver has detected a valid MFAS sequence on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where $x = 0$ to 7. The value of mfas_lock<x> is set by the 800GBASE-ER1 FEC sublayer multi-frame alignment state diagram (Figure 186–19).                                                                                                                                 | 7  |
| mfas_valid<x>                                                                                                                                                                                                                                                                                                                                                                                                    | 8  |
| A Boolean variable that is set to true if the value of the MFAS overhead described in 186.2.3.5.1 is valid on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where $x = 0$ to 7, and is set to false otherwise. The overhead is considered to be valid if the value increments in each frame, modulo 256.                                                                                                      | 9  |
| multiframe_counter_done                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
| A Boolean variable that is set to true when multiframe_counter has reached its terminal count. It is set to false when the counter is started by the 800GBASE-ER1 FEC sublayer alignment marker location state diagram (Figure 186–20).                                                                                                                                                                          | 11 |
| pma_alignment_valid                                                                                                                                                                                                                                                                                                                                                                                              | 12 |
| A Boolean variable that is set to true if the 800GBASE-ER1 PMA polarization symbol streams are aligned. 800GBASE-ER1 PMA polarization symbol streams are considered to be aligned when faw_lock<y> is true for both $y$ , each 800GBASE-ER1 PMA polarization symbol stream has a unique identifier, and the 800GBASE-ER1 PMA polarization symbol streams are deskewed. Otherwise, this variable is set to false. | 13 |
| PMA_reset                                                                                                                                                                                                                                                                                                                                                                                                        | 14 |
| Boolean variable that is true when set by a management entity and is false otherwise.                                                                                                                                                                                                                                                                                                                            | 15 |
| pma_restart_lock                                                                                                                                                                                                                                                                                                                                                                                                 | 16 |
| A Boolean variable that used to restart the FAW field lock process on both 800GBASE-ER1 PMA polarization symbol streams when 15 consecutive frame alignment word sequences fail to match on either 800GBASE-ER1 PMA polarization symbol stream. The value of pma_restart_lock is set by the 800GBASE-ER1 PMA FAW field lock state diagram (Figure 186–17).                                                       | 17 |
| pma_signal_ok                                                                                                                                                                                                                                                                                                                                                                                                    | 18 |
| A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive. It is true if the value was OK and false if the value was FAIL.                                                                                                                                                                                                                | 19 |
| pma_pss_mapping<y>                                                                                                                                                                                                                                                                                                                                                                                               | 20 |
| A variable that holds the value of the pma_pss received on the 800GBASE-ER1 PMD physical signal $y$ , where $y = 0$ to 1. The value pma_pss_mapping<y> is set by the 800GBASE-ER1 PMA FAW field lock state diagram (Figure 186–17).                                                                                                                                                                              | 21 |
| raml_align                                                                                                                                                                                                                                                                                                                                                                                                       | 22 |
| Boolean variable that indicates when the 800GBASE-R PCS alignment markers insertion location needs to be aligned to the received AML overhead. The value of raml_align is set by the 800GBASE-ER1 FEC sublayer alignment marker location state diagram (Figure 186–20).                                                                                                                                          | 23 |

|                                                                                                                                                                                                                                   |    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| raml_valid                                                                                                                                                                                                                        | 1  |
| Boolean variable that indicates if the received information in the AML field is valid. It is set by the RAML_CHECK function.                                                                                                      | 2  |
|                                                                                                                                                                                                                                   | 3  |
| reset_fec                                                                                                                                                                                                                         | 4  |
| Boolean variable that controls the resetting of the 800GBASE-ER1 FEC sublayer. It is true whenever a reset is necessary including when FEC_reset is true and during power on, and is false otherwise.                             | 5  |
|                                                                                                                                                                                                                                   | 6  |
|                                                                                                                                                                                                                                   | 7  |
|                                                                                                                                                                                                                                   | 8  |
| reset_pma                                                                                                                                                                                                                         | 9  |
| Boolean variable that controls the resetting of the 800GBASE-ER1 PMA sublayer. It is true whenever a reset is necessary including when PMA_reset is true and during power on, and is false otherwise.                             | 10 |
|                                                                                                                                                                                                                                   | 11 |
|                                                                                                                                                                                                                                   | 12 |
| sof_raml                                                                                                                                                                                                                          | 13 |
| Indicates the value of raml_counter at the start of an 800GBASE-ER1 FEC sublayer multi-frame. The value sof_raml is set by the 800GBASE-ER1 FEC sublayer alignment marker location state diagram (Figure 186–20).                 | 14 |
|                                                                                                                                                                                                                                   | 15 |
|                                                                                                                                                                                                                                   | 16 |
| test_fam<x>                                                                                                                                                                                                                       | 17 |
| A Boolean variable that is set to true when a candidate FAM field position is available for testing on 800GBASE-ER1 FEC sublayer tributary flow <i>x</i> , where <i>x</i> = 0 to 7, and false when the FIND_1ST state is entered. | 18 |
|                                                                                                                                                                                                                                   | 19 |
|                                                                                                                                                                                                                                   | 20 |
|                                                                                                                                                                                                                                   | 21 |
| test_faw<y>                                                                                                                                                                                                                       | 22 |
| A Boolean variable that is set to true when a candidate FAW field position is available for testing on 800GBASE-ER1 PMD physical signal <i>y</i> , where <i>y</i> = 0 to 1, and false when the FIND_1ST state is entered.         | 23 |
|                                                                                                                                                                                                                                   | 24 |
|                                                                                                                                                                                                                                   | 25 |
|                                                                                                                                                                                                                                   | 26 |

### 186.4.2.3 Functions

|                                                                                                                                                                                                                                                                                                                                                                                                          |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| FAM_COMPARE                                                                                                                                                                                                                                                                                                                                                                                              | 27 |
| This function compares the values of first_fam<x> and current_fam<x> to determine if a valid alignment mechanism pattern has been detected and returns the result of the comparison using the variable fam_match. If current_fam<x> and first_fam<x> found a match, fam_match is set to true, otherwise it is set to false.                                                                              | 28 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 29 |
| FAM_SLIP                                                                                                                                                                                                                                                                                                                                                                                                 | 30 |
| Causes the next candidate FAM field position to be tested. The precise method for determining the next candidate FAM field position is not specified and is implementation dependent. However, an implementation shall ensure that all possible FAM field positions are evaluated.                                                                                                                       | 31 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 32 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 33 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 34 |
| FAW_COMPARE                                                                                                                                                                                                                                                                                                                                                                                              | 35 |
| This function compares the values of first_pma_pss<y> and current_pma_pss<y> to determine if a valid frame alignment word sequence has been detected and returns the result of the comparison using the variable faw_match. If current_pma_pss<y> and first_pma_pss<y> indicate the same 800GBASE-ER1 PMA polarization symbol stream identifier, faw_match is set to true, otherwise it is set to false. | 36 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 37 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 38 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 39 |
| FAW_SLIP                                                                                                                                                                                                                                                                                                                                                                                                 | 40 |
| Causes the next candidate FAW field position to be tested. The precise method for determining the next candidate FAW field position is not specified and is implementation dependent. However, an implementation shall ensure that all possible FAW field positions are evaluated.                                                                                                                       | 41 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 42 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 43 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 44 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 45 |
| MFAS_COMPARE                                                                                                                                                                                                                                                                                                                                                                                             | 46 |
| This function compares the values of first_mfas<x> and current_mfas<x> to determine if a valid multi-frame alignment sequence has been detected and returns the result of the comparison using                                                                                                                                                                                                           | 47 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 48 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 49 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 50 |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 51 |
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|                                                                                                                                                                                                                                                                                                                                                                                                          | 54 |

the variable `mfas_valid`. If `current_mfas<x>` is one more than `first_mfas<x>`, modulo 256, `mfas_valid` is set to true.

#### RAML\_ALIGN

Updates the value of `raml_counter` based on the received AML field after compensating for the number of payload 257-bit blocks processed in the 800GBASE-ER1 FEC sublayer multi-frame

#### RAML\_CHECK

Compares the value of `sof_raml` with the value of the received AML field. If they match, `raml_valid` is set to true, otherwise `raml_valid` is set to false.

### 186.4.2.4 Counters

#### `fam_bad_count<x>`

Counts the number of consecutive 800GBASE-ER1 FEC sublayer frame alignment mechanism patterns that don't match the expected value on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where  $x = 0$  to 7.

#### `fam_counter<x>`

Counts the interval of 2560 257-bit blocks containing 800GBASE-ER1 FEC sublayer frame alignment mechanism patterns on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where  $x = 0$  to 7.

#### `faw_bad_count<y>`

Counts the number of consecutive frame alignment word sequences that don't match the expected value for a given 800GBASE-ER1 PMA polarization stream, where  $y = 0$  to 1.

#### `faw_counter<y>`

This counter counts the interval of 175 104 symbols between the starting position of one FAW field and the expected starting position of the next FAW field on a 800GBASE-ER1 PMA polarization symbol stream, where  $y = 0$  to 1. This counter starts with a value of zero when `start_faw_counter` is asserted (at the position of the first symbol of a matched frame alignment word sequence on a 800GBASE-ER1 PMA polarization symbol stream), is incremented for each symbol on that 800GBASE-ER1 PMA polarization symbol stream, and reaches a terminal value of 175 104 at the expected position of the next FAW field on that 800GBASE-ER1 PMA polarization symbol stream.

#### `frame_counter<x>`

Counts the interval of 82 240 bytes between 800GBASE-ER1 multi-frame alignment overhead on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where  $x = 0$  to 7.

#### `mfas_bad_count<x>`

Counts the number of consecutive 800GBASE-ER1 FEC sublayer multi-frame alignment overhead sequences that don't match the expected value on an 800GBASE-ER1 FEC sublayer tributary FEC flow, where  $x = 0$  to 7.

#### `multiframe_counter`

Counts the interval of 657 920 bytes between 800GBASE-ER1 AML overhead

#### `raml_bad_cnt`

Counts the number of consecutive 800GBASE-ER1 FEC sublayer multi-frames where the received AML value doesn't match the expected value

#### `raml_counter`

Counts the number of 257-bit blocks since the last alignment marker location

#### `zero_aml_cnt`

Counts the number of consecutive 800GBASE-ER1 FEC sublayer multi-frames where the received AML value is 0.

### 186.4.3 State diagram figures

The 800GBASE-ER1 PMA shall implement two FAW field lock processes as shown in Figure 186–17 to identify the boundaries of 800GBASE-ER1 PMA multi-frames by observing the stream of symbols on four physical signals. The synchronization process operates independently on each polarization. The state diagram determines when the 800GBASE-ER1 PMA has detected the location of the frame alignment word sequence in the received symbol stream for the 800GBASE-ER1 PMD service interface.

The 800GBASE-ER1 FEC sublayer shall implement eight FAM field lock processes as shown in Figure 186–18 to identify the FAM field at the start of each of the eight tributary FEC flows distributed from the 800GBASE-ER1 FEC sublayer tributary frame by observing data from the ER1 FEC decoder output after removal of CRC32 and pad fields and distribution to the eight tributary FEC flows.

The 800GBASE-ER1 FEC sublayer shall implement eight multi-frame alignment process as shown in Figure 186–19 to identify the overhead multi-frame by observing the pattern in the MFAS field of each of the eight tributary FEC flows.

If the alignment marker location feature is enabled, the 800GBASE-ER1 FEC sublayer shall implement the alignment marker location synchronization process shown in Figure 186–20. The AMs are inserted at their original position (matching the position from before AMs were removed by far-end transmit function) as indicated by the RAML value. When an unexpected RAML value arrives, the previous position of the AM is maintained until 8 consecutive unexpected RAML values are received, after which the AM position is updated to the new position indicated by the RAML.

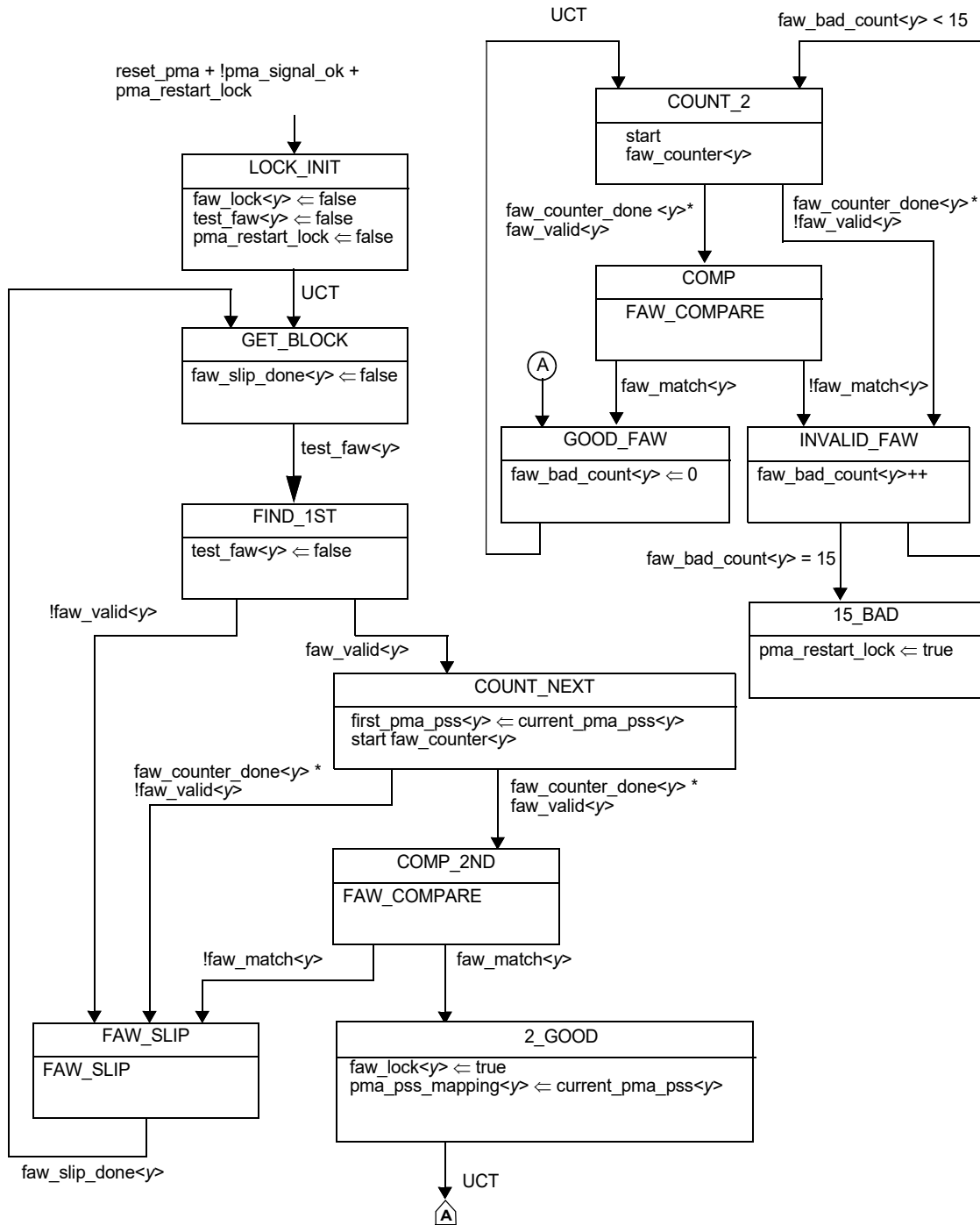


Figure 186-17—800GBASE-ER1 PMA FAW field lock state diagram

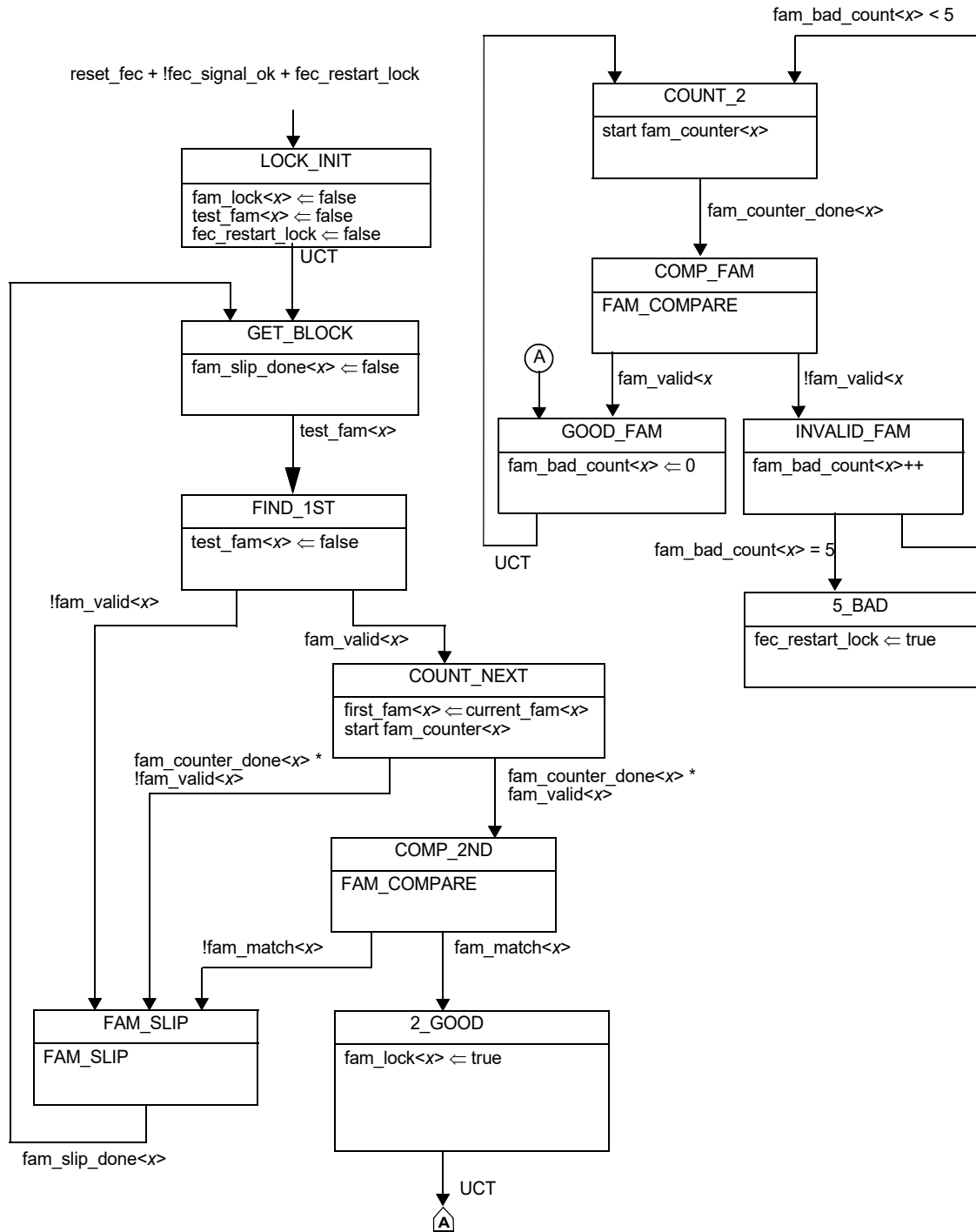


Figure 186-18—800GBASE-ER1 FEC sublayer FAM field lock state diagram

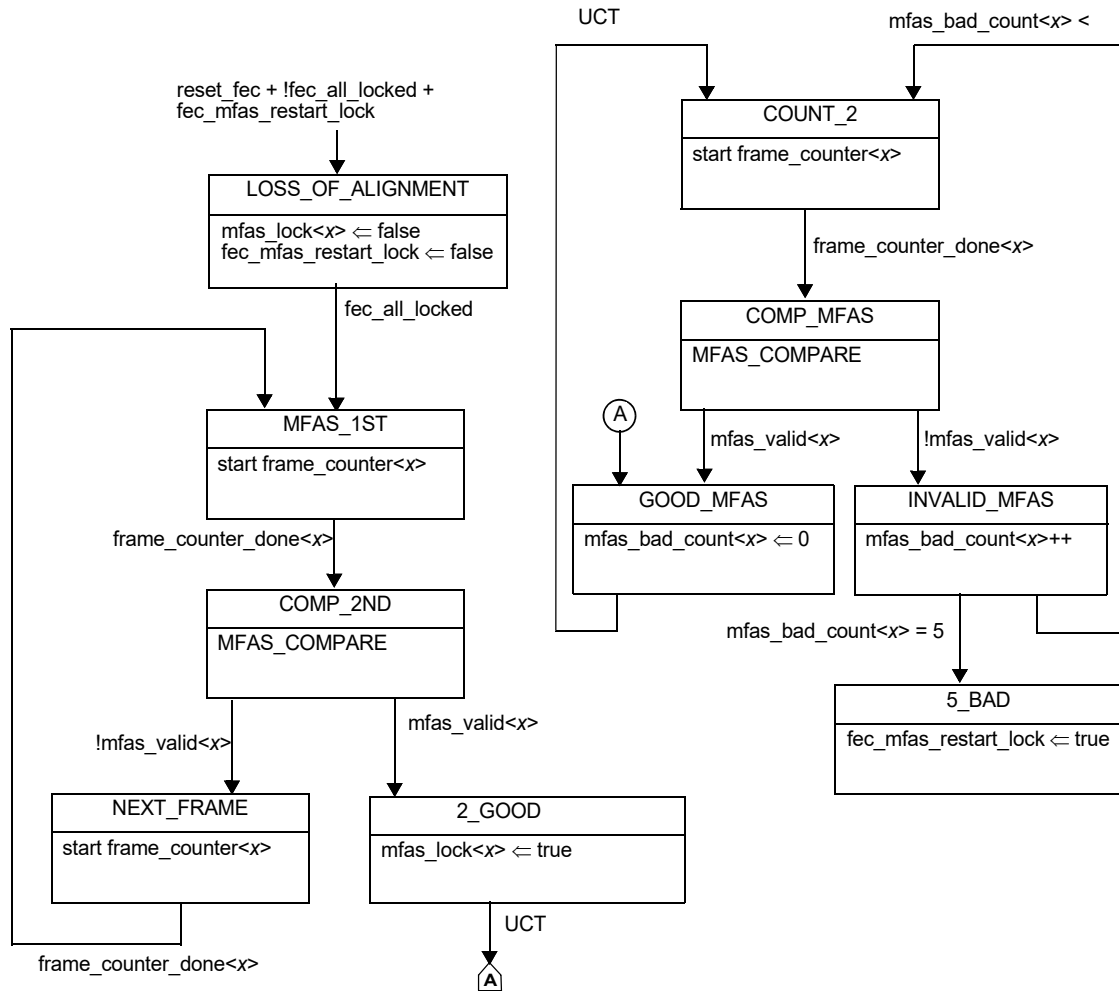


Figure 186-19—800GBASE-ER1 FEC sublayer multi-frame alignment state diagram

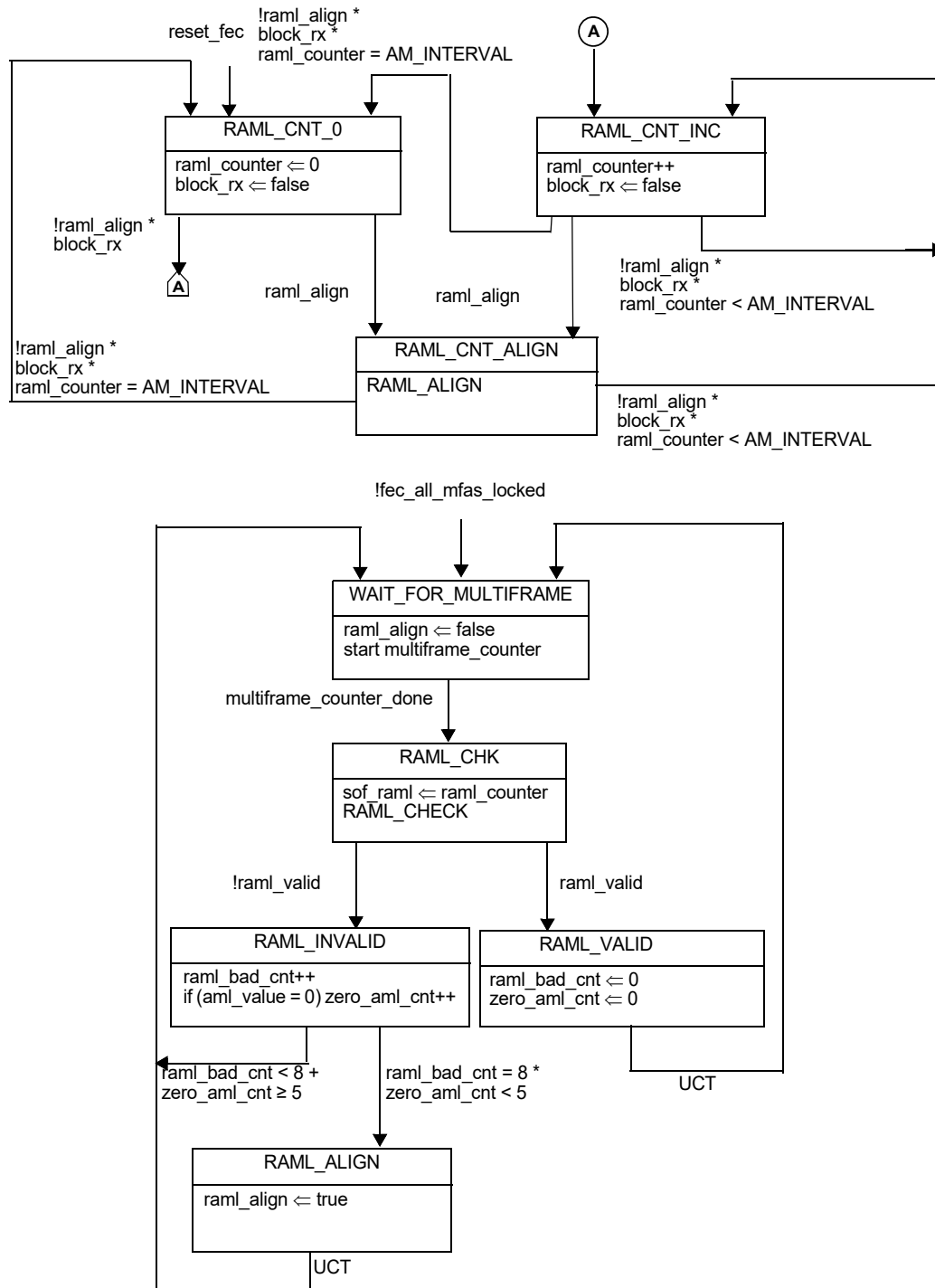


Figure 186-20—800GBASE-ER1 FEC sublayer alignment marker location state diagram

## 186.5 Delay constraints

The maximum delay contributed by the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers (sum of transmit and receive delays at one end of the link) shall be no more than 4 000 256 bit times (7813 pause\_quanta or 5000.32 ns). A description of overall system delay constraints and the definitions for bit times and pause\_quanta can be found in 169.4 and its references.

## 186.6 Path data delay for time synchronization

The 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers optionally provide path data delay information for time synchronization (see Clause 90).

### 186.6.1 800GBASE-ER1 FEC sublayer path data delay for time synchronization

The ability to report 800GBASE-ER1 FEC sublayer path data delays is indicated by the FEC\_delay\_ns\_TX\_ability, FEC\_delay\_subns\_TX\_ability, FEC\_delay\_ns\_RX\_ability, and FEC\_delay\_subns\_RX\_ability status variables (see Table 186–10).

The transmit and receive path data delay values are reported as the sum of the delays through the three sections that make up the 800GBASE-ER1 FEC sublayer (see Figure 186–3). The lane distribution and interleaving processes performed by the Inverse RS-FEC function and ER1 FEC encoder can cause significant delay variations as described in 90.7.1. As recommended in 90A.7, the maximum delay value is allocated to the transmit path and the minimum delay value to the receive path.

The delay through the Inverse RS-FEC section is reported as the delay incurred by the first bit of a set of interleaved RS-FEC codewords, ignoring any skew. The transmit path data delay is reported as if all lanes at the FEC Service Interface arrive at the same time as the last arriving lane. The delay through the Mapping/Demapping and ER1 FEC encoding/decoding sections is reported as the delay incurred by the first bit of the first non-fixed-stuff 257-bit GMP word of the tributary 0 multi-frame, where the start of the 800GBASE-ER1 tributary frame is also the start of an ER1 FEC frame.

Four separate delays are reported in the following eight path data delay status variables (see Table 186–10):

- FEC\_delay\_ns\_TX\_max, FEC\_delay\_subns\_TX\_max
- FEC\_delay\_ns\_TX\_min, FEC\_delay\_subns\_TX\_min
- FEC\_delay\_ns\_RX\_max, FEC\_delay\_subns\_RX\_max
- FEC\_delay\_ns\_RX\_min, FEC\_delay\_subns\_RX\_min

### 186.6.2 PMA path data delay for time synchronization

The ability to report PMA path data delays is indicated by the PMA\_delay\_ns\_TX\_ability, PMA\_delay\_subns\_TX\_ability, PMA\_delay\_ns\_RX\_ability, and PMA\_delay\_subns\_RX\_ability status variables (see Table 186–12).

The 800GBASE-ER1 PMA sublayer has an intrinsic delay pattern where the sum of the transmit and receive delays is a constant value. As recommended in 90A.7, the maximum delay value is allocated to the transmit path and the minimum delay value to the receive path. The transmit and receive path data delay values are reported as the delay incurred by the first bit of the first data symbol of the PMA frame S<0>.

Four separate delays are reported in the following eight path data delay status variables (see Table 186–12):

- PMA\_delay\_ns\_TX\_max, PMA\_delay\_subns\_TX\_max
- PMA\_delay\_ns\_TX\_min, PMA\_delay\_subns\_TX\_min

- PMA\_delay\_ns\_RX\_max, PMA\_delay\_subns\_RX\_max
- PMA\_delay\_ns\_RX\_min, PMA\_delay\_subns\_RX\_min

## 186.7 Management variables

### 186.7.1 800GBASE-ER1 FEC sublayer management

FEC control and status variables intended to be accessible via a management system are listed in Table 186–9 and Table 186–10.

If the MDIO Interfaces (see Clause 45) is implemented, FEC control and status variables shall be mapped to MDIO registers as shown in Table 186–9 and Table 186–10.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

**Table 186–9—800GBASE-ER1 FEC sublayer control variables and MDIO mapping**

| Control variable                       | Variable reference | MDIO register/bit number | MDIO register/bit reference |
|----------------------------------------|--------------------|--------------------------|-----------------------------|
| FEC_alignment_marker_location_enable   | 186.2.3.5.10       | 1.1825.14                | 45.2.1.177c                 |
| FEC_reset                              | 186.4.2.1          | 1.0.15                   | 45.2.1.1                    |
| FEC_rx_test_mode                       | 186.2.4.11         | 1.1501.0                 | 45.2.1.170                  |
| FEC_tx_test_mode                       | 186.2.3.12         | 1.1501.3                 | 45.2.1.170                  |
| IFEC_bypass_indication_enable          | 186.2.3.1.3        | 1.2200.1                 | 45.2.1.216                  |
| IFEC_degraded_SER_enable               | 186.2.3.1.3        | 1.2200.4                 | 45.2.1.216                  |
| IFEC_degraded_SER_activate_threshold   | 186.2.3.1.3        | 1.650, 1.651             | 45.2.1.146                  |
| IFEC_degraded_SER_deactivate_threshold | 186.2.3.1.3        | 1.652, 1.653             | 45.2.1.147                  |
| IFEC_degraded_SER_interval             | 186.2.3.1.3        | 1.654, 1.655             | 45.2.1.148                  |
| IFEC_local_loopback_enable             | 186.2.5.1          | 1.2200.5                 | 45.2.1.216                  |
| IFEC_remote_loopback_enable            | 186.2.5.2          | 1.2200.6                 | 45.2.1.216                  |

### 186.7.2 800GBASE-ER1 PMA management

PMA control and status variables intended to be accessible via a management system are listed in Table 186–9 and Table 186–10. Additional variables associated with the APSU functions are listed in Table 178B–5 and Table 178B–6.

If the MDIO Interfaces (see Clause 45) is implemented, PMA control and status variables shall be mapped to MDIO registers as shown in Table 186–9 and Table 186–10.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

**Table 186–10—800GBASE-ER1 FEC sublayer status variables and MDIO mapping**

| Status variable                       | Variable reference | MDIO register/bit number             | MDIO register/bit reference |
|---------------------------------------|--------------------|--------------------------------------|-----------------------------|
| FEC_corrected_cw_counter              | 186.2.4.1          | 1.2412 to 1.2413                     | 45.2.1.258                  |
| FEC_uncorrected_cw_counter            | 186.2.4.1          | 1.2414 to 1.2415                     | 45.2.1.259                  |
| FEC_total_bits_counter                | 186.2.4.1          | 1.2416 to 1.2419                     | 45.2.1.260                  |
| FEC_corrected_bits_counter            | 186.2.4.1          | 1.2420 to 1.2423                     | 45.2.1.261                  |
| FEC_crc_total_blocks_counter          | 186.2.4.5          | 1.2640 to 1.2642                     | 45.2.1.265                  |
| FEC_crc_errored_blocks_counter        | 186.2.4.5          | 1.2643                               | 45.2.1.266                  |
| FEC_alignment_marker_location_ability | 186.2.3.5.10       | 1.1800.8                             | 45.2.1.175                  |
| FEC_delay_ns_TX_max                   | 186.6.1            | 1.1813, 1.1814                       | 45.2.1.177a                 |
| FEC_delay_subns_TX_max                | 186.6.1            | 1.1817                               | 45.2.1.177a                 |
| FEC_delay_ns_TX_min                   | 186.6.1            | 1.1815, 1.1816                       | 45.2.1.177a                 |
| FEC_delay_subns_TX_min                | 186.6.1            | 1.1818                               | 45.2.1.177a                 |
| FEC_delay_ns_RX_max                   | 186.6.1            | 1.1819, 1.1820                       | 45.2.1.177b                 |
| FEC_delay_subns_RX_max                | 186.6.1            | 1.1823                               | 45.2.1.177b                 |
| FEC_delay_ns_RX_min                   | 186.6.1            | 1.1821, 1.1822                       | 45.2.1.177b                 |
| FEC_delay_subns_RX_min                | 186.6.1            | 1.1824                               | 45.2.1.177b                 |
| FEC_delay_ns_RX_ability               | 186.6.1            | 1.1800.4                             | 45.2.1.175                  |
| FEC_delay_ns_TX_ability               | 186.6.1            | 1.1800.5                             | 45.2.1.175                  |
| FEC_delay_subns_RX_ability            | 186.6.1            | 1.1800.6                             | 45.2.1.175                  |
| FEC_delay_subns_TX_ability            | 186.6.1            | 1.1800.7                             | 45.2.1.175                  |
| IFEC_local_loopback_ability           | 186.2.5.1          | 1.2201.6                             | 45.2.1.217.6b               |
| FEC_800G_remote_loopback_ability      | 186.2.5.2          | 1.2201.7                             | 45.2.1.217.6a               |
| IFEC_align_status                     | 186.2.3.1.1        | 1.2201.14                            | 45.2.1.217                  |
| IFEC_amps_lock<0:31>                  | 186.2.3.1.1        | 1.2201.8 to 1.2201.11                | 45.2.1.217                  |
| IFEC_cw_counter                       | 186.2.3.1.3        | 1.2207, 1.2208, 1.2209               | 45.2.1.220a                 |
| IFEC_codeword_error_bin<1:15>         | 186.2.3.1.3        | 1.2734 to 1.2763                     | 45.2.1.268                  |
| IFEC_symbol_error_counter<0:31>       | 186.2.3.1.3        | 1.2212 to 1.2241<br>1.2770 to 1.2799 | 45.2.1.222                  |

**Table 186–10—800GBASE-ER1 FEC sublayer status variables and MDIO mapping (contin-**

| Status variable                | Variable reference | MDIO register/bit number | MDIO register/bit reference |
|--------------------------------|--------------------|--------------------------|-----------------------------|
| IFEC_pcs_lane_mapping<0:31>    | 186.2.3.1.3        | 1.2700 to 1.2731         | 45.2.1.268, 45.2.1.269      |
| IFEC_bypass_indication_ability | 186.2.3.1.3        | 1.2201.1                 | 45.2.1.217                  |
| IFEC_degraded_SER_ability      | 186.2.3.1.3        | 1.2201.3                 | 45.2.1.217                  |
| IFEC_degraded_SER              | 186.2.3.1.3        | 1.2201.12                | 45.2.1.217                  |
| IFEC_hi_ser                    | 186.2.3.1.3        | 1.2201.2                 | 45.2.1.217                  |
| IFEC_remote_phy_fault          | 186.2.3.5.2        | 1.2411.2                 | 45.2.1.257                  |
| IFEC_rx_rm_degraded            | 186.2.3.1.3        | 1.2201.4                 | 45.2.1.217                  |
| IFEC_rx_local_degraded         | 186.2.3.1.3        | 1.2201.5                 | 45.2.1.217                  |
| IFEC_corrected_cw_counter      | 186.2.3.1.3        | 1.2202, 1.2203           | 45.2.1.218                  |
| IFEC_uncorrected_cw_counter    | 186.2.3.1.3        | 1.2204, 1.2205           | 45.2.1.219                  |

**Table 186–11—PMA control variables and MDIO mapping**

| Control variable | Variable reference | MDIO register/bit number | MDIO register/bit reference |
|------------------|--------------------|--------------------------|-----------------------------|
| PMA_reset        | 186.4.2.1          | 1.0.15                   | 45.2.1.1                    |

**Table 186–12—PMA status variables and MDIO mapping**

| Status variable         | Variable reference | MDIO register/bit number | MDIO register/bit reference |
|-------------------------|--------------------|--------------------------|-----------------------------|
| PMA_delay_ns_TX_max     | 186.6.2            | 1.1801, 1.1802           | 45.2.1.176                  |
| PMA_delay_subns_TX_max  | 186.6.2            | 1.1809                   | 45.2.1.176                  |
| PMA_delay_ns_TX_min     | 186.6.2            | 1.1803, 1.1804           | 45.2.1.176                  |
| PMA_delay_subns_TX_min  | 186.6.2            | 1.1810                   | 45.2.1.176                  |
| PMA_delay_ns_RX_max     | 186.6.2            | 1.1805, 1.1806           | 45.2.1.177                  |
| PMA_delay_subns_RX_max  | 186.6.2            | 1.1811                   | 45.2.1.177                  |
| PMA_delay_ns_RX_min     | 186.6.2            | 1.1807, 1.1808           | 45.2.1.177                  |
| PMA_delay_subns_RX_min  | 186.6.2            | 1.1812                   | 45.2.1.177                  |
| PMA_delay_ns_RX_ability | 186.6.2            | 1.1800.0                 | 45.2.1.175                  |
| PMA_delay_ns_TX_ability | 186.6.2            | 1.1800.1                 | 45.2.1.175                  |

**Table 186–12—PMA status variables and MDIO mapping (*continued*)**

| Status variable            | Variable reference | MDIO register/bit number | MDIO register/bit reference |
|----------------------------|--------------------|--------------------------|-----------------------------|
| PMA_delay_subns_RX_ability | 186.6.2            | 1.1800.2                 | 45.2.1.175                  |
| PMA_delay_subns_TX_ability | 186.6.2            | 1.1800.3                 | 45.2.1.175                  |

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## 186.8 Protocol implementation conformance statement (PICS) proforma for Clause 186, Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer type 800GBASE-ER1

### 186.8.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 186, Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer type 800GBASE-ER1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 186.8.2 Identification

#### 186.8.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                                         |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                                                                                                                                                                                   |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                                                                                                                                                                                 |  |
| Implementation Name(s) and Version(s) <sup>1,3</sup>                                                                                                                                                                                                                                                    |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup>                                                                                                                                                     |  |
| <p>NOTE 1—Required for all implementations.</p> <p>NOTE 2—May be completed as appropriate in meeting the requirements for the identification.</p> <p>NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).</p> |  |

#### 186.8.2.2 Protocol summary

|                                                                                                                                                                               |                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                           | IEEE Std 802.3dj-202x, Clause 186, Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer type 800GBASE-ER1 |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                               |                                                                                                                                            |
| <p>Have any Exception items been required? No [ ] Yes [ ]</p> <p>(See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)</p> |                                                                                                                                            |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

### 186.8.3 PICS proforma tables for Forward Error Correction sublayer (FEC), type 800GBASE-ER1

#### 186.8.3.1 Major capabilities/options

| Item | Feature                                       | Subclause                        | Value/Comment                                           | Status | Support           |
|------|-----------------------------------------------|----------------------------------|---------------------------------------------------------|--------|-------------------|
| *MD  | MDIO                                          | 186.7.1                          | Registers and interface supported                       | O      | Yes [ ]<br>No [ ] |
| DC   | Delay constraints                             | 186.5                            | No more than 4 000 256 BT or 7813 pause quanta or       | M      | Yes [ ]           |
| *LBL | Support for ER1 FEC local loopback            | 186.2.5.1                        | Supports local loopback                                 | O      | Yes [ ]<br>No [ ] |
| *LBR | Support for ER1 FEC remote loopback           | 186.2.5.2                        | Supports remote loopback                                | O      | Yes [ ]<br>No [ ] |
| *AML | Support for ER1 FEC alignment marker location | 186.2.3.5.1<br>0,<br>186.2.4.7.5 | Supports preserving alignment marker location           | O      | Yes [ ]<br>No [ ] |
| *TS  | Support for time synchronization              | 186.6.1                          | Supports reporting path delays for time synchronization | O      | Yes [ ]<br>No [ ] |

#### 186.8.3.2 Transmit function

| Item | Feature                           | Subclause                              | Value/Comment | Status | Support           |
|------|-----------------------------------|----------------------------------------|---------------|--------|-------------------|
| TF1  | Inverse RS-FEC                    | 186.2.3.1                              |               | M      | Yes [ ]           |
| TF2  | 257-bit block distribution        | 186.2.3.2                              |               | M      | Yes [ ]           |
| TF3  | GMP mapper                        | 186.2.3.3                              |               | M      | Yes [ ]           |
| TF4  | FAM and pad insertion             | 186.2.3.4                              |               | M      | Yes [ ]           |
| TF5  | General overhead insertion        | 186.2.3.5.1<br>through<br>186.2.3.5.11 |               | M      | Yes [ ]           |
| TF6  | Alignment marker location feature | 186.2.3.5.10                           |               | AML:M  | Yes [ ]<br>No [ ] |
| TF7  | 128-bit interleaving              | 186.2.3.6                              |               | M      | Yes [ ]           |
| TF8  | CRC32                             | 186.2.3.7                              |               | M      | Yes [ ]           |
| TF9  | ER1 FEC frame                     | 186.2.3.8                              |               | M      | Yes [ ]           |
| TF10 | Scrambler                         | 186.2.3.9                              |               | M      | Yes [ ]           |
| TF11 | ER1 FEC encoding and interleaving | 186.2.3.10                             |               | M      | Yes [ ]           |
| TF12 | Transmission bit ordering         | 186.2.3.11                             |               | M      | Yes [ ]           |
| TF13 | Test pattern generator            | 186.2.3.12                             |               | M      | Yes [ ]           |

### 186.8.3.3 Receive function

| Item | Feature                           | Subclause                       | Value/Comment | Status | Support           |
|------|-----------------------------------|---------------------------------|---------------|--------|-------------------|
| RF1  | ER1 FEC deinterleave and decode   | 186.2.4.1                       |               | M      | Yes [ ]           |
| RF2  | Descrambler                       | 186.2.4.3                       |               | M      | Yes [ ]           |
| RF3  | Pad removal                       | 186.2.4.4                       |               | M      | Yes [ ]           |
| RF4  | CRC32 check and error marking     | 186.2.4.5                       |               | M      | Yes [ ]           |
| RF5  | 128-bit distribution              | 186.2.4.6                       |               | M      | Yes [ ]           |
| RF6  | FAM and general OH fields         | 186.2.4.7.1 through 186.2.4.7.7 |               | M      | Yes [ ]           |
| RF7  | GMP demapper                      | 186.2.4.8                       |               | M      | Yes [ ]           |
| RF8  | 257-bit collection                | 186.2.4.9                       |               | M      | Yes [ ]           |
| RF9  | Inverse RS-FEC                    | 186.2.4.10                      |               | M      | Yes [ ]           |
| RF10 | Alignment marker location feature | 186.2.4.10.3                    |               | AML:M  | Yes [ ]<br>No [ ] |

### 186.8.3.4 Time synchronization

| Item | Feature                   | Subclause | Value/Comment                                                   | Status | Support                      |
|------|---------------------------|-----------|-----------------------------------------------------------------|--------|------------------------------|
| TS1  | Path data delays          | 186.6.1   | Path data delays reported using DDMP as defined in 186.6.1      | TS:M   | Yes [ ]<br>N/A [ ]           |
| TS2  | Nanosecond resolution     | 186.6.1   | Path data delays reported with nanosecond or better resolution. | TS:M   | Yes [ ]<br>N/A [ ]           |
| TS3  | Sub-nanosecond resolution | 186.6.1   | Path data delays reported with sub-nanosecond resolution.       | TS:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |

### 186.8.3.5 Loopbacks

| Item | Feature         | Subclause | Value/Comment                   | Status | Support            |
|------|-----------------|-----------|---------------------------------|--------|--------------------|
| LB1  | Local loopback  | 186.2.5.1 | Meets requirements in 186.2.5.1 | LBL:M  | Yes [ ]<br>N/A [ ] |
| LB2  | Remote loopback | 186.2.5.2 | Meets requirements in 186.2.5.2 | LBR:M  | Yes [ ]<br>N/A [ ] |

### 186.8.3.6 Management

| Item | Feature                                              | Subclause | Value/Comment | Status | Support            |
|------|------------------------------------------------------|-----------|---------------|--------|--------------------|
| M1   | Alternate access to management variables is provided | 186.7.1   |               | !MD:M  | Yes [ ]<br>N/A [ ] |
| M2   | Mapping of MDIO control bits and MDIO status bits    | 186.7.1   |               | MD:M   | Yes [ ]<br>N/A [ ] |

### 186.8.4 PICS proforma tables for Physical Media Attachment (PMA) sublayer, type 800GBASE-ER1

#### 186.8.4.1 Major capabilities/options

| Item | Feature                          | Subclause | Value/Comment                                           | Status | Support           |
|------|----------------------------------|-----------|---------------------------------------------------------|--------|-------------------|
| *MD  | MDIO                             | 186.7.2   | Registers and interface supported                       | O      | Yes [ ]<br>No [ ] |
| DC   | Delay constraints                | 186.5     | No more than 4 000 256 BT or 7813 pause quanta or       | M      | Yes [ ]           |
| *TS  | Support for time synchronization | 186.6.2   | Supports reporting path delays for time synchronization | O      | Yes [ ]<br>No [ ] |

#### 186.8.4.2 Transmit function

| Item | Feature                                       | Subclause | Value/Comment | Status | Support |
|------|-----------------------------------------------|-----------|---------------|--------|---------|
| TF1  | Gray mapping and polarization distribution    | 186.3.3.1 |               | M      | Yes [ ] |
| TF2  | PMA frame and multiframe                      | 186.3.3.2 |               | M      | Yes [ ] |
| TF3  | Pilot sequence                                | 186.3.3.3 |               | M      | Yes [ ] |
| TF4  | Training sequence                             | 186.3.3.4 |               | M      | Yes [ ] |
| TF5  | Multi-frame alignment word and Reserved field | 186.3.3.5 |               | M      | Yes [ ] |
| TF6  | 16QAM encode                                  | 186.3.3.6 |               | M      | Yes [ ] |
| TF8  | Transmit bit ordering                         | 186.3.3.7 |               | M      | Yes [ ] |

### 186.8.4.3 Receive function

| Item | Feature                                          | Subclause | Value/Comment | Status | Support |
|------|--------------------------------------------------|-----------|---------------|--------|---------|
| RF1  | Receive signal sampling                          | 186.3.4.1 |               | M      | Yes [ ] |
| RF2  | Receive signal processing                        | 186.3.4.2 |               | M      | Yes [ ] |
| RF3  | FAW, TS, Reserved, and PS removal                | 186.3.4.3 |               | M      | Yes [ ] |
| RF4  | Polarization combining and symbol deinterleaving | 186.3.4.4 |               | M      | Yes [ ] |
| RF5  | Signal indication logic                          | 186.3.4.5 |               | M      | Yes [ ] |

### 186.8.4.4 Time synchronization

| Item | Feature                   | Subclause | Value/Comment                                                   | Status | Support                      |
|------|---------------------------|-----------|-----------------------------------------------------------------|--------|------------------------------|
| TS1  | Path data delays          | 186.6.2   | Path data delays reported using DDMP as defined in 186.6.2      | TS:M   | Yes [ ]<br>N/A [ ]           |
| TS2  | Nanosecond resolution     | 186.6.2   | Path data delays reported with nanosecond or better resolution. | TS:M   | Yes [ ]<br>N/A [ ]           |
| TS3  | Sub-nanosecond resolution | 186.6.2   | Path data delays reported with sub-nanosecond resolution.       | TS:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |

### 186.8.4.5 Management

| Item | Feature                                                 | Subclause | Value/Comment | Status | Support            |
|------|---------------------------------------------------------|-----------|---------------|--------|--------------------|
| M1   | Equivalent access to PMA management objects is provided | 186.7.2   |               | !MD:M  | Yes [ ]<br>N/A [ ] |
| M2   | Mapping of MDIO control bits and MDIO status bits       | 186.7.2   |               | MD:M   | Yes [ ]<br>N/A [ ] |